

Keep the Angle

A Geometry-Preserving Basis in Spectral Embeddings

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Abstract

The commute-time (resistance) embedding built from the low eigenvectors of a graph Laplacian is a workhorse of spectral geometry, yet its distances are known to *degenerate* on large graphs [16]. We show the geometry does not vanish; it is largely retained by the *angular* coordinate. Writing each node’s low-mode embedding in polar form (magnitude $\times$ direction), the direction preserves graph geodesics at Spearman $\rho \approx 0.93$ on a reference manifold, while the magnitude preserves $\rho \approx 0.03$ and a random-mode basis of equal size preserves $\rho \approx 0$. Keeping only the angle is the row-normalization of Ng–Jordan–Weiss spectral clustering [8] applied to the commute-weighted ($1/\sqrt{\lambda_k}$) embedding (the weighting matters: §5.6), whose existing theory concerns cluster separation [14]; we give it a geodesic justification. We show, as a corollary of von Luxburg–Radl–Hein, that the radial coordinate of the full embedding degenerates to a local-degree quantity for intrinsic dimension $d \geq 3$ ($d = 2$ is the recurrent borderline, treated empirically), and **conjecture**, with strong empirical support, that the angular coordinate remains rank-faithful to geodesic distance—flat in both mode-count and graph size where the raw embedding decays; a conditional local form of the conjecture is proved in full, extends globally under an explicit injectivity hypothesis, and is *unconditional* on the flat torus (Theorem 8, Corollaries 9–10). The basis is **substrate-robust**: across random-geometric lattices, manifold-embedding samples, and Wolfram-model hypergraph rewriting it preserves geodesics wherever the graph carries genuine low-dimensional Riemannian geometry, and it degrades on Lorentzian causal sets and geometry-destroying small-world graphs. Angular fidelity falls roughly linearly with intrinsic dimension, and is invariant to sampling density only after a Coifman–Lafon $\alpha=1$ normalization (§5.8). The result gives Ng–Jordan–Weiss row-normalization a geodesic interpretation complementary to its clustering theory.

1 Introduction

Spectral embeddings—Laplacian eigenmaps [1], diffusion and commute-time maps [5]—represent a graph by the low eigenvectors of its Laplacian and underpin much of manifold learning and spectral clustering. They carry a well-known pathology: on large geometric graphs the commute (resistance) distance degenerates to a function of local degrees alone, losing all global geometry [16]. Yet the most successful spectral-clustering algorithm [8] *row-normalizes* the embedding—projects each node to the unit sphere—and works well. The existing theory of that normalization concerns cluster geometry: it explains why row-normalized embeddings of well-separated clusters concentrate in nearly orthogonal directions on the sphere [14, 15]. It does not address metric geometry. **This paper asks what row-normalization recovers of the graph’s geodesic structure, and why.**

Our answer is a clean decomposition. Write each node’s low-mode embedding in polar form, magnitude $\times$ direction. The *magnitude* is exactly the degenerate radial coordinate that von Luxburg’s

theorem kills; the *direction* carries the graph’s geodesic geometry (after the $\alpha=1$ density normalization where sampling is strongly non-uniform, §5.8). Row-normalization keeps the direction and discards the magnitude—so beyond its cluster-separation role it is also the geodesically correct projection, and its fidelity is flat in both mode-count and graph size precisely where the raw embedding decays.

The underlying operation—*separate scale from direction, then keep the part that carries the task geometry*—recurs beyond clustering: embedding compression stores the norm and quantizes the unit direction, and polar-coordinate KV-cache quantization splits key vectors the same way [12]. The recurrence is evidence for the *decomposition*, not for always discarding the radius, and the boundary is sharp in practice: for attention *keys*, per-vector angular quantization can score well on reconstruction metrics yet destroy generation, because the attention geometry $QK^\top$ depends on per-channel scale—there, scale is signal, not nuisance [10]. Our hypotheses state the boundary exactly: discarding the magnitude is the right quotient when the downstream metric satisfies the tangential-noncollapse condition (A2) of Theorem 6, i.e. when the discarded radial variable is nuisance structure for the metric being preserved. For graph geodesics the radius is degree/density noise (Corollary 1) and the quotient is justified; for KV keys it is not. It also admits a physical reading: in observer theory a bounded observer perceives geometry only after coarse-graining a discrete substrate [19], and the angular projection is a candidate for that coarse-graining. We keep that interpretation to the discussion (§7) and let the spectral result stand on its own; the empirical core is that the basis is *substrate-robust*: it preserved geodesics on every graph family we tested that carries genuine low-dimensional Riemannian geometry, independent of how it was generated. (An exploratory probe of the same diagnostic on a real semantic embedding space—a language model’s moral-text vectors—appears in Appendix D.)

Contributions: (1) the **Keep-the-Angle Principle** (§3)—a Radial-Degeneracy *corollary* (for the full embedding, from [16]) with a spectral-gap bound on the truncation error (Proposition 2) and an empirically-supported Angular-Preservation *conjecture* now with first metric-distortion evidence, a fully proved conditional local form (Theorem 8, Appendix B) resting on a general deterministic angular-transfer theorem for arbitrary filtered spectral embeddings (Theorem 6, Corollary 7), unconditional on the flat torus (Corollary 10), and an explicit distortion-to-rank bridge (Proposition 11); (2) a measurement with controls (§5), a weighting ablation (§5.6), and a bake-off against named spectral distances in which the angle beats every purely spectral competitor (§5.7); (3) substrate robustness across five substrate families with a two-factor small-world screen (§5.3); (4) a fidelity-vs-dimension scaling law (§5.4); (5) an exploratory cross-domain probe on a language-model semantic embedding (Appendix D); and (6) two honest negatives (§6). We do *not* claim to derive physics; the core result is a basis-level fact about spectral embeddings.

2 Background

Spectral geometry. Laplacian eigenmaps [1] and diffusion maps [5] embed a graph via low eigenvectors; the commute-time embedding scales mode k by $1/\sqrt{\lambda_k}$. **Commute-time degeneracy.** von Luxburg et al. [16] proved the commute distance of a large geometric graph degenerates, $R(i, j) \rightarrow 1/k_i + 1/k_j$ (k_i the degree of node i); we read this backwards as the collapse of the *radial* coordinate to local degree. They also propose an *amplified* commute distance as a correction, the natural competitor to ours; rather than repair the radius we discard it and keep the angle. **Row-normalization.** Ng–Jordan–Weiss [8] row-normalize the eigen-embedding—the angular projection we study. Schiebinger, Wainwright and Yu [14] give the sharpest existing account: under a mixture model, row-normalized embeddings concentrate clusters in orthogonal directions,

explaining k -means’ post-normalization success; consistency of spectral clustering is treated in [15]. Both analyses are about cluster separation on the sphere. Our question—whether the normalized embedding preserves *geodesic distances* on a manifold-like graph—is complementary, and to our knowledge open. The same normalization removes degree as a nuisance in degree-corrected block models—SCORE [29] cancels it through eigenvector ratios, and spherical spectral clustering under the DCBM [30] projects to the sphere—but those are community-recovery guarantees, not geodesic ones. The embedding Ψ itself is Rustamov’s Global Point Signature [28] from shape analysis, and the magnitude we discard is used in its own right for cluster and anomaly detection [31]; we keep the direction and give it a metric reading. **Emergent geometry and observer theory.** In the Wolfram model [17, 18, 6] space is a hypergraph and law is what a bounded observer perceives after coarse-graining the ruliad [19]; that coarse-graining is left qualitative, and our angular projection is a candidate for it (§7). **Hyperbolic embedding.** Trees embed in $\mathbb{H}^2$ at low distortion [7, 13, 9]; we test and reject a hyperbolic reframe in §6.

3 The Keep-the-Angle Principle

Let G be connected on n nodes with symmetric-normalized Laplacian $L = I - D^{-1/2}AD^{-1/2}$, eigenpairs (λ_k, v_k) , $0 = \lambda_0 < \lambda_1 \leq \dots$. The low-mode spectral embedding using the lowest m nontrivial modes places node i at $\Psi_i = (v_1(i)/\sqrt{\lambda_1}, \dots, v_m(i)/\sqrt{\lambda_m})$. Write $\Psi_i = r_i \hat{u}_i$ with radius $r_i = \|\Psi_i\|$ and angle $\hat{u}_i = \Psi_i/\|\Psi_i\|$. Throughout, “angular distance” means the Euclidean *chordal* distance between unit vectors, $\|\hat{u}_i - \hat{u}_j\| = 2\sin(\theta_{ij}/2)$ with θ_{ij} the angle between $\hat{u}_i$ and $\hat{u}_j$; it is monotone in θ_{ij} , so all rank (Spearman) statements are unchanged by the chordal-vs-arc choice, and the figures use the chordal form.¹

Intrinsic dimension and degree notation. Throughout, d is the intrinsic dimension of the substrate, and k_i denotes the degree of node i (we reserve d for dimension to avoid the collision d vs. d_i): the manifold dimension *by construction* for the random-geometric benchmarks (tori, king-lattices), and for emergent graphs the value at which the three independent estimators of §4 (ball-growth, spectral dimension, effective rank) mutually agree—reported with their inter-estimator spread, with only low-spread graphs treated as manifolds. Dimension regimes use this measured intrinsic dimension, with one convention enforced throughout: the radial-degeneracy *theorem* is gated at $d \geq 3$; $d = 2$ is the recurrent borderline, where the degeneracy appears with logarithmic corrections and is treated empirically (§5.5); and in (quasi-)1D the radius stays geodesic-informative, as predicted (§5.4). The angular *effect* is observed from quasi-1D upward.

Corollary 1 (Radial Degeneracy). *Let G_n be random geometric graphs of intrinsic dimension $d \geq 3$ satisfying the conditions of [16] (the $d = 2$ case is marginal: at the recurrent boundary the resistance carries logarithmic corrections, matching the borderline behaviour we measure in §5.5), and let Φ_i be the full commute-time embedding built from the unnormalized Laplacian, so that $\|\Phi_i\|^2 = L_{ii}^+$. Assume additionally that the degree ratio $\max_i k_i / \min_i k_i$ stays bounded (as it does with high probability in the graph models of [16]). As $n \rightarrow \infty$ the resistance distance degenerates uniformly and multiplicatively, $\sup_{i \neq j} |R(i, j)/(1/k_i + 1/k_j) - 1| \rightarrow 0$ [16], and this pairwise statement transfers to the diagonal: $L_{ii}^+ = (1/k_i)(1 + o(1))$ uniformly over nodes, so the radius satisfies $\|\Phi_i\| \rightarrow 1/\sqrt{k_i}$. The radius of the full commute embedding is therefore asymptotically geometry-free—a local-density*

¹We call Ψ “commute-style” rather than the commute-time embedding proper: the classical commute embedding uses the pseudoinverse of the *unnormalized* Laplacian and all $n - 1$ modes, so that $\|\Psi_i - \Psi_j\|^2$ equals the resistance $R(i, j)$ exactly. Our Ψ is the truncated, normalized analogue standard in spectral clustering; the distinction matters for Corollary 1 and is handled there.

coordinate carrying no geodesic information. (The diagonal does not follow from the pairwise limits alone, which determine only combinations $L_{ii}^+ + L_{jj}^+ - 2L_{ij}^+$; the gauge $\sum_j L_{ij}^+ = 0$ and a trace identity close the gap, with the degree-ratio bound controlling the error propagation. Full proof: Appendix A.)

Remark (Truncated, normalized radius). Our experiments use the truncated radius $r_i = \|\Psi_i\|$ (m lowest modes of L_{sym}), not $\|\Phi_i\|$. Transferring Corollary 1 to this setting needs (i) the normalized analogue of the degeneracy [16] and (ii) control of the truncation error. We settle (ii) unconditionally below and support (i) numerically; the residual gap is then the angular claim itself (Conjecture 3). Numerically the truncated radius tracks the local degree, $\rho(r_i, 1/\sqrt{k_i}) = 0.79\text{--}0.87$ across $m = 5\text{--}40$, while the full normalized radius is near-constant (coefficient of variation 0.09) and radius-only geodesic preservation is $\rho \leq 0.08$ (§5.5).

Proposition 2 (Truncation is spectral-gap bounded). *Let $\Psi_i^\infty = (v_k(i)/\sqrt{\lambda_k})_{k=1}^{n-1}$ be the full symmetric-normalized commute-style embedding and Ψ_i^m its truncation to the lowest m nontrivial modes. Then for all nodes $i \neq j$,*

$$0 \leq \|\Psi_i^\infty\|^2 - \|\Psi_i^m\|^2 \leq \frac{1}{\lambda_{m+1}}, \quad 0 \leq \|\Psi_i^\infty - \Psi_j^\infty\|^2 - \|\Psi_i^m - \Psi_j^m\|^2 \leq \frac{2}{\lambda_{m+1}}.$$

Proof. Each left inequality is termwise: truncation deletes only the summands with $k > m$, of the form $v_k(i)^2/\lambda_k \geq 0$ or $(v_k(i) - v_k(j))^2/\lambda_k \geq 0$. For the right inequalities, $\lambda_k \geq \lambda_{m+1}$ for every $k > m$, so the deleted mass is at most $\lambda_{m+1}^{-1} \sum_{k>m} v_k(i)^2$ and $\lambda_{m+1}^{-1} \sum_{k>m} (v_k(i) - v_k(j))^2$ respectively. The eigenvectors $\{v_k\}_{k=0}^{n-1}$ are orthonormal, so $VV^\top = I$ gives $\sum_k v_k(i)^2 = 1$ and, for $i \neq j$, $\sum_k v_k(i)v_k(j) = 0$; hence $\sum_{k>m} v_k(i)^2 \leq 1$ and $\sum_{k>m} (v_k(i) - v_k(j))^2 \leq \sum_k (v_k(i) - v_k(j))^2 = \sum_k v_k(i)^2 + \sum_k v_k(j)^2 - 2\sum_k v_k(i)v_k(j) = 2$. These identity sums run over all $k = 0, \dots, n-1$, including the trivial mode $k = 0$ that Ψ omits; it sits only in the subtracted low-mode partial sums, and dropping its nonnegative terms can only tighten the tail bounds. $\square$

Two consequences. First, the error is controlled by the inverse spectral gap $1/\lambda_{m+1}$ alone. This is *not* an n -independent absolute bound: on a fixed-degree neighbourhood graph refined toward the continuum λ_{m+1} itself shrinks, so $1/\lambda_{m+1}$ grows. What it controls is the truncation error *relative to the embedding scale*, which is fixed by the same low eigenvalues; since the angular and rank statistics we report are invariant to global scale, that relative control is what transfers, and is why the continuum refinement of §5.10 does not degrade. Second, the deleted content is high-frequency (large- λ); the geometry lives in the retained low modes (an equal-size random-mode basis preserves $\rho \approx 0$, §5). But this bound is a *sanity check, not the angular theorem*: it controls the full-vs-truncated discrepancy at a fixed graph and mode cutoff, in a norm weighted by inverse eigenvalues; it does not show the deleted tail is irrelevant to angular *rank* geometry, and it gives no uniform-in- n angular stability statement. The empirical flatness in m (§5.5) remains *evidence for* Conjecture 3, not a consequence of Proposition 2. On the 2-torus ($n = 2000$) the bound holds with room to spare: the realized radius error is 3.5–17% of $1/\lambda_{m+1}$ across $m \in \{5, 10, 20, 40\}$, the full normalized radius is near-constant (coefficient of variation 0.09), and the truncated radius stays rank-correlated with $1/\sqrt{k_i}$. What remains open is not the radius but the angle: Proposition 2 bounds how far truncation moves the metric, not that the surviving *angular* coordinate equals the geodesic—that is Conjecture 3.

Conjecture 3 (Angular Preservation). *For ε -graph or k -NN samples of a compact Riemannian manifold satisfying the density conditions of [3, 23] (an explicit class, not one defined post hoc by our own estimators), row-normalization $\hat{u}_i = \Psi_i/\|\Psi_i\|$ deletes the degenerate density factor of Corollary 1, and the pairwise distances among the resulting directions remain rank-faithful to*

geodesic distance—Spearman correlation bounded away from zero (empirically ≈ 0.9)—uniformly in the mode-count m and size n . (We reserve “rank-faithful” for $\text{Spearman} \geq \rho_0 > 0$ over the stated class; the diagnostic reading—using the ω -screen and dimension spread to decide whether a given graph lies in the class—is a separate corollary-in-use, kept out of the conjecture to avoid circularity.) The empirical level ≈ 0.9 is uniform in sampling density only after the Coifman–Lafon $\alpha=1$ normalization (§5.8); under strongly non-uniform sampling the raw L_{sym} angle stays rank-faithful (Spearman bounded away from zero) but its level drifts with density contrast, so the ≈ 0.9 clause is asserted for near-uniform density or, generally, the $\alpha=1$ angle. A stronger, metric form of the conjecture—bi-Lipschitz to geodesic distance with constants uniform in m and n at fixed scale—is now known to be **false** for the commute weighting: high modes raise the local angular speed like $\sqrt{\Lambda}$ (Proposition 13; exactly computable on S^1), so the metric strengthening must be either scale-dependent—constants uniform in m for pairs above the spectral wavelength $\Lambda^{-1/2}$, which remains open—or filtered: for the heat/diffusion weighting the uniform-in-truncation local bi-Lipschitz statement is a theorem in the continuum (Theorem 12). The rank clause above is untouched by this: the empirical flatness in m is rank stability under a fixed observer budget, not an m -independent fixed-scale metric claim, and our evidence is rank-based.

Metric evidence on the torus. The metric form is no longer entirely untested. On the 2-torus we compare embedded distance to graph geodesic directly. After a single global scale, the angular map has a robust empirical bi-Lipschitz constant $\kappa \approx 2.6$ (ratio of the 97.5th to 2.5th percentile of $d_{\text{emb}}/d_{\text{geo}}$ over 19,900 pairs), tighter than the full commute embedding (3.1), an equal-size random-mode basis (8.3), or the magnitude coordinate (125); and its leading four coordinates Procrustes-match the canonical flat-torus embedding in $\mathbb{R}^4$ at disparity 0.07, versus 0.995 for random modes. This is genuine metric distortion, not rank correlation—one substrate, but a well-understood one. (The full-range constant is inflated by a few near-antipodal pairs, where the chord/arc ratio dips; the map stays injective—nothing is folded together—so we report the percentile-robust κ .) The right uniform metric target is scale-dependent for the commute filter (Proposition 13): fixed-scale constants uniform in m are impossible, while uniformity above the spectral wavelength is open—this measurement is one datapoint at the tested budget.

3.1 Toward a proof of Conjecture 3

Proposition 2 controls truncation; the remaining question is whether row-normalization preserves the geometry the low modes carry. It does, and the mechanism is an exact identity. To set expectations plainly: the payoff of this subsection is a *transfer* theorem, not a proof of the conjecture—it reduces the conjecture, at fixed m , to two continuum hypotheses (local eigenmap bi-Lipschitzness, H1, and radius-flat transversality, H2), plus an injectivity hypothesis (H3) for the global form.

Lemma 4 (Normalization preserves bi-Lipschitz geometry). *For nonzero $u, v \in \mathbb{R}^m$ with $\hat{u} = u/\|u\|$,*

$$\|\hat{u} - \hat{v}\|^2 = \frac{\|u - v\|^2 - (\|u\| - \|v\|)^2}{\|u\| \|v\|}.$$

If $F : (X, d) \rightarrow \mathbb{R}^m$ satisfies $L^{-1}d(x, y) \leq \|F(x) - F(y)\| \leq Ld(x, y)$ with $\rho^- \leq \|F\| \leq \rho^+$ and radius Λ -Lipschitz, then $\hat{F} = F/\|F\|$ obeys $\|\hat{F}(x) - \hat{F}(y)\| \leq (L/\rho^-)d(x, y)$, and if $\Lambda < 1/L$,

$$\|\hat{F}(x) - \hat{F}(y)\| \geq \frac{\sqrt{L^{-2} - \Lambda^2}}{\rho^+} d(x, y).$$

Proof. The identity combines $\|\hat{u}-\hat{v}\|^2 = 2-2\langle u, v \rangle / (\|u\|\|v\|)$ with $\langle u, v \rangle = \frac{1}{2}(\|u\|^2 + \|v\|^2 - \|u-v\|^2)$. The upper bound uses numerator $\leq \|u-v\|^2 \leq L^2 d^2$ and $\|u\|\|v\| \geq (\rho^-)^2$. For the lower bound, $(\|u\| - \|v\|)^2 \leq \Lambda^2 d^2$ and $\|u-v\|^2 \geq d^2/L^2$ give numerator $\geq d^2(L^{-2} - \Lambda^2)$; divide by $\|u\|\|v\| \leq (\rho^+)^2$. $\square$

The condition $\Lambda < 1/L$ is a quantitative radial degeneracy: the radius must vary *slower* than the embedding moves. As $\Lambda \rightarrow 0$ (radius $\rightarrow$ constant, the limit of Corollary 1) the lower constant tends to $1/(L\rho^+)$ and normalization becomes a pure scaling. On the 2-torus this holds pointwise: the identity is exact to 10^{-15} , and the radius Lipschitz constant $\Lambda = 0.0918$ sits only marginally under the map's lower stretch $A = 0.0920$. The margin is razor-thin, so Lemma 4's *worst-case* lower constant $\sqrt{L^{-2} - \Lambda^2}/\rho^+$ is near-vacuous here—but that crude bound is not the operative one. Proposition 5 shows the lower stretch is the pointwise *tangential* component of dF , which stays bounded away from zero (0/9200 local pairs are radially degenerate, median Λ five times below A), giving a measured local angular bi-Lipschitz constant of 2.2 that is an empirical fact, not a consequence of the crude lemma. Theorem 6 below turns this observation into the hypothesis itself: the transfer runs on two-point tangential noncollapse (A2), and the crude $\Lambda < 1/L$ enters only as one sufficient way to verify it—so the razor-thin radius margin on the torus costs the theory nothing.

Proposition 5 (Exact angular distortion). *Let F be differentiable with $r = \|F\| > 0$ and $\hat{F} = F/r$. Taking $y \rightarrow x$ in Lemma 4 gives, for every tangent vector v ,*

$$\|d\hat{F}(v)\|^2 = \frac{\|dF(v)\|^2 - dr(v)^2}{r^2} = \frac{\|dF(v) - \hat{u} dr(v)\|^2}{r^2}, \quad dr(v) = \langle dF(v), \hat{u} \rangle.$$

The angular stretch in a direction v is *exactly* the tangential part of $dF(v)$ —the component orthogonal to the radius—divided by r . So $\hat{F}$ is a local bi-Lipschitz immersion *iff* dF is nowhere purely radial ($dF(v) \nparallel F$ for $v \neq 0$), the lower stretch being $\inf_v \|dF(v) - \hat{u} dr(v)\|/(r\|v\|)$. This sharpens Lemma 4: the crude $\Lambda < 1/L$ merely forces the radial term $dr(v)$ below the total $\|dF(v)\|$, whereas the proposition shows what is really required is that the embedding's velocity never aligns with its position. Lemma 4 and Proposition 5 link the two halves of the principle and reduce the conjecture to standard ingredients.

The general transfer principle. Proposition 5 identifies the mechanism infinitesimally; its finite-distance form can be made the *hypothesis* itself. The result is a transfer theorem that knows nothing about manifolds, Laplacians, or eigenvectors—and of which the eigenmap theorem below (Theorem 8) is the commute-weighted special case. Any uniformly calibrated discrete embedding inherits local angular bi-Lipschitz geometry from a continuum embedding whenever the continuum displacement has a two-point tangential component bounded below; radius flatness is only one sufficient route to that tangential noncollapse.

Theorem 6 (Deterministic angular transfer). *Let (X, d) be a metric space, $\{x_1, \dots, x_n\} \subset X$, and $F : X \rightarrow \mathbb{R}^m \setminus \{0\}$ with radius $R = \|F\|$ and angular map $N = F/R$. Let $Z_{n,i} \in \mathbb{R}^m \setminus \{0\}$ be any discrete embedding of x_i , with angular rows $\hat{Z}_{n,i} = Z_{n,i}/\|Z_{n,i}\|$, and fix a pair domain $\mathcal{P} \subset X \times X$ (e.g. $\{0 < d(x, y) \leq r_0\}$). Assume: (A1) $0 < \rho^- \leq R \leq \rho^+ < \infty$ on the relevant region; (A2) **two-point tangential noncollapse**: there are $0 < \tau \leq \beta < \infty$ with*

$$\tau d(x, y) \leq \left(\|F(x) - F(y)\|^2 - (R(x) - R(y))^2 \right)^{1/2} \leq \beta d(x, y) \quad \text{for all } (x, y) \in \mathcal{P}$$

(the middle quantity is exactly the part of the displacement that survives row-normalization; infinitesimally it is the tangential component of Proposition 5); (A3) uniform calibration: there are a

global scale $s_n > 0$, an orthogonal $Q_n \in O(m)$, and $E_n \rightarrow 0$ with $\max_{i \leq n} \|Q_n(s_n Z_{n,i}) - F(x_i)\| \leq E_n$. Then for every $\eta \in (0, 1)$, once $E_n < \rho^-/2$, every sampled pair in $\mathcal{P}$ with

$$d(x_i, x_j) \geq \theta_n(\eta) := \frac{4E_n \rho^+}{\eta \tau \rho^-} \rightarrow 0$$

obeys

$$(1 - \eta) \frac{\tau}{\rho^+} d(x_i, x_j) \leq \|\hat{Z}_{n,i} - \hat{Z}_{n,j}\| \leq \left(\frac{\beta}{\rho^-} + \eta \frac{\tau}{\rho^+} \right) d(x_i, x_j).$$

Proof. The identity of Lemma 4 with $u = F(x)$, $v = F(y)$ reads $\|N(x) - N(y)\|^2 = [\|F(x) - F(y)\|^2 - (R(x) - R(y))^2] / (R(x)R(y))$, so (A1)–(A2) make N bi-Lipschitz on $\mathcal{P}$ with constants $a = \tau/\rho^+$ and $b = \beta/\rho^-$. Pairwise angular distances of $\hat{Z}$ are invariant under the scale s_n and the orthogonal Q_n , and by (A3) with normalization stability (Lemma 16, using $\|F(x_i)\| \geq \rho^-$ in the denominator) each calibrated angular row lies within $2E_n/\rho^-$ of $N(x_i)$; hence each pairwise angular distance lies within $4E_n/\rho^-$ of the continuum one. For $d(x_i, x_j) \geq \theta_n(\eta)$ that error is at most $\eta a d(x_i, x_j)$; adding and subtracting it from the continuum bounds gives the claim. $\square$

Corollary 7 (Filtered spectral embeddings). *Let M and m (at a spectral gap) be as in Theorem 8, and let h be a positive Lipschitz spectral filter on the retained window ($H_m := \max_{k \leq m} h(\mu_k)$, L_h a Lipschitz constant). Define the filtered eigenmap $F_h(x) = (h(\mu_k) \varphi_k(x))_{k=1}^m$ and graph rows $Z_{n,i}^{(h)} = (h(\tilde{\lambda}_k) v_k(i))_{k=1}^m$ with $\tilde{\lambda}_k = \lambda_k/c_n$. Under the sup-norm spectral consistency of Appendix B (Lemma 17 with $\text{diag}(\mu_k^{-1/2})$ replaced by $\text{diag}(h(\mu_k))$; same proof), hypothesis (A3) holds with high probability with*

$$E_{n,h} = H_m \epsilon_n + L_h \delta_n \sqrt{m} (\Phi_m + \epsilon_n) \rightarrow 0.$$

Hence if F_h satisfies (A1)–(A2) on $\{d_M \leq r_0\}$, then with high probability the angular map of the h -filtered graph embedding is bi-Lipschitz to d_M , with the constants of Theorem 6, for all sampled pairs with $\theta_n(\eta) \leq d_M \leq r_0$. The commute filter $h(\lambda) = \lambda^{-1/2}$ recovers Theorem 8; diffusion $h(\lambda) = e^{-t\lambda}$ and plain $h \equiv 1$ are equally admissible at fixed m . The weighting ablation of §5.6—plain eigenmaps collapse as m grows while commute and diffusion stay flat—is therefore a statement about how each filter’s constants $(\tau, \beta, \rho^\pm)$ behave in m , not about the transfer mechanism, which is filter-independent.

Theorem 8 (Conditional angular transfer for the commute-weighted eigenmap; local form of Conjecture 3). *Let M be a compact d -manifold whose graph Laplacian spectrally converges to M ’s ([3] for ε -graphs, [23] for k -NN graphs), and fix a mode-count m at a spectral gap of M . Assume (H1) the truncated eigenmap $F : x \mapsto (\varphi_k(x)/\sqrt{\mu_k})_{k=1}^m$ is L_0 -bi-Lipschitz onto its image on geodesic balls of radius r_0 —the quantitative truncated-eigenmap embedding property of [20, 21, 22]—and (H2) the associated truncated radius is Λ_0 -Lipschitz with $\Lambda_0 < 1/L_0$ and lies in $[\rho^-, \rho^+]$. Set*

$$A = \frac{\sqrt{L_0^{-2} - \Lambda_0^2}}{2\rho^+}, \quad B = \frac{2L_0}{\rho^-},$$

both independent of n . Then there is a resolution scale $\theta_n \rightarrow 0$ (explicitly, a constant multiple of the sup-norm spectral-consistency rate of [23]) such that, with high probability as $n \rightarrow \infty$, the graph angular map obeys $A d_M(x, y) \leq \|\hat{u}_x - \hat{u}_y\| \leq B d_M(x, y)$ for all sampled x, y with $\theta_n \leq d_M(x, y) \leq r_0$.

Proof (outline; complete proof in Appendix B). This is the special case of Theorem 6 and Corollary 7 with $h(\lambda) = \lambda^{-1/2}$ and $\eta = 1/2$: (H1)–(H2) imply the tangential noncollapse (A2) with $\tau = \sqrt{L_0^{-2} - \Lambda_0^2}$ (positive because $\Lambda_0 < 1/L_0$) and $\beta = L_0$, giving $A = (1 - \eta)\tau/\rho^+$ and $B \geq \beta/\rho^- + \eta\tau/\rho^+$. Note (H2) is only one sufficient route: the theorem survives with (H2) replaced by (A2) directly, which is strictly weaker and is the condition the torus actually satisfies with a healthy margin (the discussion after Lemma 4). Self-contained proof: by (H1), (H2) and Lemma 4, the *continuum* angular map $N = F/\|F\|$ is bi-Lipschitz on r_0 -balls with constants $a = \sqrt{L_0^{-2} - \Lambda_0^2}/\rho^+$ and $b = L_0/\rho^-$ (Lemma 15). The transfer to the graph needs *sup-norm* eigenvector convergence—the ∞ -norm rates of [23] rather than generic L^2 consistency—so that normalization survives where $\|F\|$ is smallest: after the graph-to-continuum eigenvalue scaling and a block-orthogonal alignment across the multiplets below the gap (both invisible to pairwise angular distances), the graph embedding is calibrated to F within an error $E_n \rightarrow 0$ *uniformly over nodes* (Lemma 17); in particular the graph radius inherits $[\rho^- - E_n, \rho^+ + E_n]$, and the graph angle lands within $2E_n/\rho^-$ of N (Lemma 16). The two-sided bound then follows with $A = a/2$ and $B = b + a/2 \leq 2L_0/\rho^-$ for every sampled pair above the scale $\theta_n = 8E_n/(a\rho^-)$. The cutoff is necessary, not an artifact: below the graph’s resolution, combinatorially twin nodes receive identical entries in every eigenvector below the localized twin-difference mode—in particular in every low mode the angular map uses—so $\hat{u}_i = \hat{u}_j$ while $d_M > 0$ (Appendix B). $\square$

From local to global. Theorem 8 is local because (H1) is. One further hypothesis—the global sibling of the radial-fold transversality of Proposition 5—globalizes it.

Corollary 9 (Global conditional form). *Assume (H1)–(H2) and additionally (H3): the continuum angular map $N = F/\|F\|$ is injective on M . Let $c_0 := \min\{\|N(x) - N(y)\| : d_M(x, y) \geq r_0\}$, which is positive by (H3) and compactness. Then, with*

$$A_g = \min\left(A, \frac{c_0}{2 \operatorname{diam} M}\right), \quad B_g = \max\left(B, \frac{2}{r_0}\right),$$

with high probability as $n \rightarrow \infty$ the graph angular map obeys $A_g d_M(x, y) \leq \|\hat{u}_x - \hat{u}_y\| \leq B_g d_M(x, y)$ for all sampled x, y with $d_M(x, y) \geq \theta_n$ (proof: Appendix B, Step 5).

(H3) is genuinely extra: Portegies-style mode counts [22] make F itself injective, but $N(x) = N(y)$ requires only that F *radially align* two points ($F(x) = tF(y)$, $t > 0$), which injectivity of F does not exclude. Empirically it holds on the torus—the angular map folds nothing together (metric-evidence paragraph above). On the flat torus, in fact, nothing at all is left to assume:

Corollary 10 (Unconditional instance: the flat torus). *Let $M = \mathbb{R}^2/(2\pi\mathbb{Z})^2$ with $m = 4$ (the full first multiplet, a spectral gap). Then (H1)–(H3) hold, with explicit constants: the truncated eigenmap is the Clifford embedding $F(x, y) = \sqrt{2}(\cos x, \sin x, \cos y, \sin y)$, which is L_0 -bi-Lipschitz globally ($r_0 = \operatorname{diam} M$) with $L_0 = \sqrt{2}$; the radius is constant, $R \equiv 2$, so $\Lambda_0 = 0$ and $\rho^\pm = 2$; and $N = F/2$ is injective. Consequently, with high probability as $n \rightarrow \infty$, the graph angular map of an ε -graph or k -NN sample of the flat torus is globally bi-Lipschitz to geodesic distance above the resolution scale, unconditionally, with $A = 1/(4\sqrt{2})$ and $B = \sqrt{2}$; the sharp continuum angular distortion is exactly $\pi/2$ (the chord-vs-arc ratio $2\sin(\delta/2)/\delta \in [2/\pi, 1]$ per coordinate—the near-antipodal dip of the metric-evidence paragraph). The statement is basis-independent: any L^2 -orthonormal basis of the four-dimensional eigenspace is an $O(4)$ rotation of the Clifford one, invisible to angular distances (proof: Appendix B, Step 6).*

The measured graph-level $\kappa \approx 2.6$ (at $m=10$, inside the second multiplet) sits above the $\pi/2 \approx 1.57$ continuum floor by finite- n and truncation slack; $m=4$ is the theoretically clean budget.

From metric to rank. The conjecture’s currency is rank (Spearman); the theorems’ is distortion. The bridge is elementary and deterministic.

Proposition 11 (Distortion controls rank). *Let $\mathcal{P}$ be a finite set of node pairs on which $A' d_M \leq \widehat{D} \leq B' d_M$ holds for some embedded distance $\widehat{D}$, write $\kappa = B'/A'$, and let $\hat{\mu}(\kappa)$ be the fraction of unordered pairs $\{P, Q\} \subset \mathcal{P}$ whose geodesic distances lie within ratio κ of each other ($\max/\min < \kappa$). Then, absent ties (ties, e.g. from integer hop distances, enter only through the standard tie corrections), Kendall’s $\tau(\widehat{D}, d_M) \geq 1 - 2\hat{\mu}(\kappa)$ and Spearman’s $\rho_S \geq 1 - 3\hat{\mu}(\kappa)$ up to an $O(1/|\mathcal{P}|)$ finite-sample correction.*

Proof. A discordant pair-of-pairs ($d_P < d_Q$ but $\widehat{D}_P > \widehat{D}_Q$) forces $A'd_Q \leq \widehat{D}_Q < \widehat{D}_P \leq B'd_P$, i.e. $d_Q/d_P < \kappa$; so the discordant fraction is at most $\hat{\mu}(\kappa)$ and $\tau = 1 - 2(\text{discordant fraction}) \geq 1 - 2\hat{\mu}(\kappa)$. Daniels’ inequality $|3\tau - 2\rho_S| \leq 1$ [25] converts this to $\rho_S \geq (3\tau - 1)/2 \geq 1 - 3\hat{\mu}(\kappa)$. $\square$

Combined with Corollary 9, this reduces the rank clause of Conjecture 3, at fixed m , to (H1)–(H3) *plus* a distance-ratio condition on the substrate: $\rho_0 = 1 - 3\mu(\kappa) > 0$ whenever $\mu(\kappa) < 1/3$ (sub- θ_n pairs are a vanishing fraction and enter the bound as $o(1)$). The reduction is honest about its weakness: it is worst-case over *all* κ -distorted maps. For the small-ball distance law $f(r) \propto r^{d-1}$ on $[0, r_0]$ an exact computation (Appendix B, Step 7) gives $\mu(\kappa) = 1 - \kappa^{-d}$, so the certificate $\tau \geq 2\kappa^{-d} - 1$ is non-vacuous only for $\kappa < 2^{1/d}$: a fixed distortion budget certifies *exponentially* less rank structure as the intrinsic dimension grows, and on substrates whose pairwise distances concentrate—the high- d regime—no finite distortion certifies any rank fidelity at all. This is a genuine ceiling for distortion-based guarantees, and a second, rank-specific mechanism for the fidelity-vs-dimension decay of §5.4. That the measured decay is *linear* ($-0.157d$, §5.4), far milder than the worst case, says the eigenmap angle is far from the worst κ -distorted map: the empirical ≈ 0.9 levels remain stronger than what distortion alone implies, which is why that clause of the conjecture stays empirical.

Uniformity in the truncation: what is true, and what is false. Everything so far is at fixed m . The uniformity-in- m question splits by filter, and both halves can now be answered in the continuum. Throughout, the cutoff is spectral and multiplet-closed: the map retains all eigenspaces with $\mu_k \leq \Lambda$, so $m = m(\Lambda)$, matching the spectral-gap convention.

Theorem 12 (Uniform-in-truncation angular embedding: heat filter). *Let M be compact, smooth, without boundary, and for $t > 0$ let $F_{t,m}(x) = (e^{-t\mu_k/2}\varphi_k(x))_{k=1}^m$ be the heat-filtered eigenmap, with $N_{t,m} = F_{t,m}/\|F_{t,m}\|$. There is $t_0(M) > 0$ such that for every fixed $t \leq t_0$ there exist $m_0(t)$, $r_t > 0$, and $0 < a_t \leq b_t < \infty$ with*

$$a_t d_M(x, y) \leq \|N_{t,m}(x) - N_{t,m}(y)\| \leq b_t d_M(x, y) \quad \text{whenever } d_M(x, y) \leq r_t,$$

for every $m \geq m_0(t)$: the constants do not deteriorate as $m \rightarrow \infty$. (Proof: Appendix C, by heat-kernel diagonal asymptotics and C^2 tail control; the exponential filter is what makes the tail summable against the polynomial growth of eigenfunction norms.)

Proposition 13 (Commute cutoff: tangential noncollapse, scale-dependent metric). *Let $F_\Lambda(x) = (\mu_k^{-1/2}\varphi_k(x))_{0 < \mu_k \leq \Lambda}$ be the commute-weighted spectral cutoff, $R_\Lambda = \|F_\Lambda\|$, $N_\Lambda = F_\Lambda/R_\Lambda$. There are $c > 0$ and Λ_0 such that for all $\Lambda \geq \Lambda_0$, uniformly over $x \in M$ and unit tangent v ,*

$$(\|dF_\Lambda(v)\|^2 - dR_\Lambda(v)^2)^{1/2} \geq c\Lambda^{d/4} :$$

the tangential component never collapses—it grows with the cutoff. Meanwhile $R_\Lambda \asymp \Lambda^{d/4-1/2}$ for $d > 2$ (and $\asymp \sqrt{\log \Lambda}$ for $d = 2$), so the angular derivative grows, $\|dN_\Lambda\| \gtrsim \Lambda^{1/2}$ ($d > 2$; $\sqrt{\Lambda/\log \Lambda}$, $d = 2$). Consequently no m -independent fixed-scale upper Lipschitz constant exists for the commute angular map: its natural resolution scale is the spectral wavelength $\Lambda^{-1/2}$. On S^1 this is exact: with all modes to frequency N , R_N is constant, $\|dF_N\|^2 = N$, and $\|dN_N\| \asymp \sqrt{N}$. (Proof: Appendix C, by the differentiated pointwise Weyl law [26, 27].)

The two statements split the uniformity question cleanly. For the *heat/diffusion* filter—one of the two weightings the ablation of §5.6 finds flat in m —uniform-in-truncation local bi-Lipschitzness is a continuum theorem. For the *commute* filter the tangential mechanism survives uniformly, with room to spare, but the metric constants are intrinsically scale-dependent: high modes raise the local angular speed like $\sqrt{\Lambda}$, so the correct uniform metric object is bi-Lipschitzness *above the wavelength scale* $\Lambda^{-1/2}$ —equivalently, a mode budget matched to the scale of interest, which is the observer-budget phenomenon of §5.10 handed a mechanism. The empirical “flat in m ” of §5.5 is thereby correctly read as *rank* stability under a fixed observer budget, not as an m -independent fixed-scale metric theorem. One gap remains at this point: the $\sqrt{\Lambda}$ angular-speed growth of Proposition 13 is filter-generic (the plain filter has the same order), so it explains the budget phenomenon but not the ablation’s dramatic *rank* separation between plain and commute (§5.6). The next proposition closes that gap.

Proposition 14 (Plain filter: distances concentrate, rank discrimination collapses). *Let $F_\Lambda^{(1)}(x) = (\varphi_k(x))_{0 < \mu_k \leq \Lambda}$ be the unweighted (plain-eigenmap) cutoff with angular map $N_\Lambda^{(1)}$, and fix $\delta > 0$. Then, uniformly over pairs with $d_M(x, y) \geq \delta$,*

$$\|N_\Lambda^{(1)}(x) - N_\Lambda^{(1)}(y)\|^2 = 2 + O(\Lambda^{-1/2}) :$$

all separated plain angular distances concentrate at the orthogonality value $\sqrt{2}$, and the vanishing correction is $-2E_\Lambda(x, y)/E_\Lambda(x, x)$ —the off-diagonal spectral function, an oscillatory kernel with sign changes on the wavelength scale and no monotone relation to d_M —so the ordering of plain angular distances carries no leading-order geodesic information. For the commute filter the same expansion gives

$$\|N_\Lambda(x) - N_\Lambda(y)\|^2 = 2 - \frac{2G_\Lambda(x, y)}{G_\Lambda(x, x)}(1 + o(1)),$$

whose correction kernel $G_\Lambda(x, y) = \sum_{0 < \mu_k \leq \Lambda} \mu_k^{-1} \varphi_k(x) \varphi_k(y)$ is low-frequency dominated (for $d \leq 3$ it converges off the diagonal to the Green’s function)—a coherent, distance-structured kernel, so the ordering of commute angular distances survives the concentration. (Proof: Appendix C.)

The mechanism in one line: the $1/\mu$ commute weighting is a low-pass filter on the *correction* kernel that carries the ranks, while equal weighting hands the correction to the oscillatory top of the spectral window. Rank statistics are invariant to the common concentration and read only that correction—which is why the ablation separates the filters at the rank level (0.06 vs 0.89 at $m=20$, §5.6) even though their crude metric orders coincide. Together, Proposition 13 explains the budget phenomenon and Proposition 14 the weighting ablation: the theory now covers both empirical laws of §5.6 and §5.10. For graph transfer at growing $m = m(n)$, the calibration of Lemma 17 extends whenever the spectral-window condition $\mathcal{E}_n(\mu_m) \gamma_m^{-1} \rightarrow 0$ holds (γ_m the gap isolating the retained block, $\mathcal{E}_n(\mu_m)$ the graph-consistency error at frequency μ_m , which grows polynomially in μ_m), and a $C^{0,1}$ graph-eigenvector theory [24] is the route to shrink the resolution cutoff from the spectral-rate scale toward the connectivity scale—optimal up to the twin-node obstruction (Appendix C).

What remains is delimited. **(H1)** is subtle: Jones–Maggioni–Schul [4] guarantee bi-Lipschitz charts only from a *well-chosen* d -subset of eigenfunctions per ball, because the lowest modes can degenerate at critical points; the truncated-eigenmap results [20, 21, 22] give quantitative control, and the constants’ uniformity in m is now split by filter: *proven* for the heat weighting (Theorem 12), *false at fixed scale* for the commute weighting (Proposition 13), open for commute in the scale-dependent form above the spectral wavelength—so the empirical “flat in m ” (§5.5) is rank stability at a fixed budget, exactly as the conjecture now states. Low-spectrum multiplicity compounds this—the square torus’s first nontrivial eigenvalue has multiplicity four, so a truncation *inside* a multiplet ($m = 5$) is basis-dependent; distances are well-defined only for whole eigenspaces, so m should be taken at a spectral gap (the choice $m=4$ of Corollary 10), or the map phrased through spectral projectors. **(H2)** is, by Proposition 5, sharper than radius flatness: what is needed is that the eigenmap’s velocity $dF(v)$ is nowhere purely radial—a transversality the immersion (H1) makes generic, and the truncated analogue of von Luxburg–Radl–Hein; Theorem 6’s (A2) is its two-point form, and is the hypothesis one should verify in practice. It holds on the torus (tangential lower stretch bounded, local constant 2.2); whether the eigenmap image ever develops a radial fold on some manifold is the sharp remaining question. Truncated eigenmaps of a general manifold are guaranteed only to be *locally* injective—(H1) supplies bi-Lipschitz charts on geodesic balls, whereas global injectivity of F requires Portegies-style mode counts [22] and injectivity of the *angular* map is stronger still—so the global metric statement is carried by the explicit hypothesis (H3) of Corollary 9. This is a limitation of general local hypotheses, not of the torus: there (H1) holds *globally* through the Clifford embedding into $\mathbb{R}^4$ and Corollary 10 is unconditional—exactly as our own Procrustes match (disparity 0.07) and finite empirical $\kappa \approx 2.6$ anticipated. Conjecture 3 is thereby reduced, in metric form, to a local-chart hypothesis (H1), a radius-flatness hypothesis (H2), and a no-radial-alignment hypothesis (H3), none of which mentions the angle directly—and, in rank form, further to a distance-ratio condition on the substrate (Proposition 11).

Status. Empirically strong (§5) but **not proven**. Its conditional local form is now a theorem with a complete proof (Theorem 8, Appendix B), globalized under (H3) (Corollary 9) and unconditional on the flat torus (Corollary 10); with Proposition 11 the rank clause itself reduces, at fixed m , to (H1)–(H3) plus a distance-ratio condition; and the uniformity-in- m question is settled in the continuum by filter—proven for heat (Theorem 12), false at fixed scale for commute (Proposition 13), with the scale-dependent commute form open. What rests on hypothesis is thus delimited to (H1)–(H3) at the tested filter, the scale-dependent commute uniformity, and the gap between worst-case and measured rank levels; the honest gap is that Corollary 1 explains why the radius *dies* for $d \geq 2$, not by itself why the angle *lives*—indeed the angular fidelity persists ($\rho \approx 0.88$ – 0.90) even on quasi-1D graphs where Corollary 1 does not apply, so the angular half is carried by the eigenmap-embedding mechanism, not degeneracy alone. That mechanism is analyzable rather than mysterious: if the low L_{sym} eigenvectors converge to Laplace–Beltrami eigenfunctions [2, 3] and eigenfunction coordinates furnish locally bi-Lipschitz charts of the underlying manifold [4], the angular map’s geodesic fidelity should follow from the chart geometry—the natural route to a proof of Conjecture 3, and this paper’s central open problem. Together these are the **Keep-the-Angle Principle**: keep $\hat{u}_i$, discard r_i — and, under strongly non-uniform sampling, keep the $\alpha=1$ density-normalized angle (§5.8).

Dimension gating. Corollary 1 is a $d \geq 3$ theorem; $d = 2$ is the recurrent borderline, where the degeneracy appears with logarithmic corrections and we treat it empirically (§5.5). In (quasi-)1D, resistance equals path length, so the radius *stays* geodesic-informative and Corollary 1 correctly does not apply—a prediction, not a failure, confirmed in §5.5.

4 Methods

Estimators. Ball-growth $N(r) \sim r^d$; spectral dimension from the heat-kernel return probability $P(t) = \langle e^{-Lt} \rangle \sim t^{-d_s/2}$; effective rank via local-PCA participation ratio. Low inter-estimator spread $\Rightarrow$ clean manifold, and the agreed value is our measured intrinsic dimension d . **Angular metric.** Spearman ρ between embedded pairwise distances and true graph geodesics for: *angle* (chordal distances between unit-normalized low-mode embeddings, §3), *random* (m random modes; control), *magnitude* (radius only), *Isomap* (classical MDS of the graph geodesics; the table of §5.3 reports it at $\dim=2$, column $\text{Isomap}_{d=2}$). **Uncertainty.** Each ρ is a rank correlation over $\binom{n_{\text{anchor}}}{2}$ node pairs ($n_{\text{anchor}} = 150$ for the substrate benchmarks, 200 for the torus metric tests (§3.1) and the cross-domain probe of Appendix D—11,175 and 19,900 pairs respectively). Bootstrap 95% confidence intervals over pair resampling are tight ($\leq \pm 0.005$ on the headline numbers), but pairs share anchor nodes and are not independent, so we treat pair-resampling CIs as optimistic floors; the dominant source of variation is the graph draw itself, so for headline ρ we also report spread over independent draws (2-torus, $m=10$: 0.914 ± 0.009 over five draws). The reference torus therefore appears at slightly different angle- ρ across tables (0.89–0.93): mode-count, anchor count, and the graph draw itself vary by section and dominate at this precision. Each table uses its section’s protocol— $m=20$, $n_{\text{anchor}}=150$ unless stated (core table §5 sweeps $m=5$ –40; the bake-off §5.7 uses $n_{\text{anchor}}=250$). **Substrates.** Arity-2/3 hypergraph rewriting (clique-expanded), plus tori, king-lattices, Poisson causal sets, and swiss-roll embedding samples. The swiss-roll case is a synthetic 2D manifold—intrinsic coordinates (u, v) , u the roll angle and v the height—carried by a random orthogonal map into $\mathbb{R}^{12}$ with isotropic noise ($\sigma=0.03$) and connected by a $k=10$ nearest-neighbour graph on $n=2000$ ambient points. It stands in for sequential/embedding-derived data (each item a point on a low- d semantic manifold, consecutive items nearby), not a specific text corpus. The Lorentzian causal set is an $n=2000$ Poisson sprinkling in a 1+1D Minkowski diamond (light-cone coordinates u, v uniform on the unit square); $p \prec q$ iff $u_p < u_q$ and $v_p < v_q$, and each point is linked to its $k=6$ nearest causal-*past* neighbours in proper time $\tau^2 = \Delta t^2 - \Delta x^2$, symmetrized for the undirected geodesic BFS—so edges encode short causal *relations* (nearest past neighbours in proper time, which need not be Hasse links), and the intrinsic $d=2$ is the sprinkling dimension.

5 Results

5.1 The core result

Estimators recover ground-truth dimension within ≈ 0.3 . On a 2-torus the polar decomposition of the low-mode embedding (Fig. 1) gives:

m	full (mag. $\times$ dir.)	angle only	magnitude only
5	0.761	0.926	0.025
10	0.650	0.913	0.035
20	0.550	0.891	0.028
40	0.517	0.928	0.034

The angle carries the geometry (≈ 0.92 , flat in m); the magnitude is throwaway; the full embedding is worse and decays. A random-mode basis reconstructs geodesics at $\rho \approx 0$ (Fig. 2); averaged over ten random draws $\rho = 0.08 \pm 0.16$, individual draws occasionally catching a low mode and spiking to ~ 0.4 .

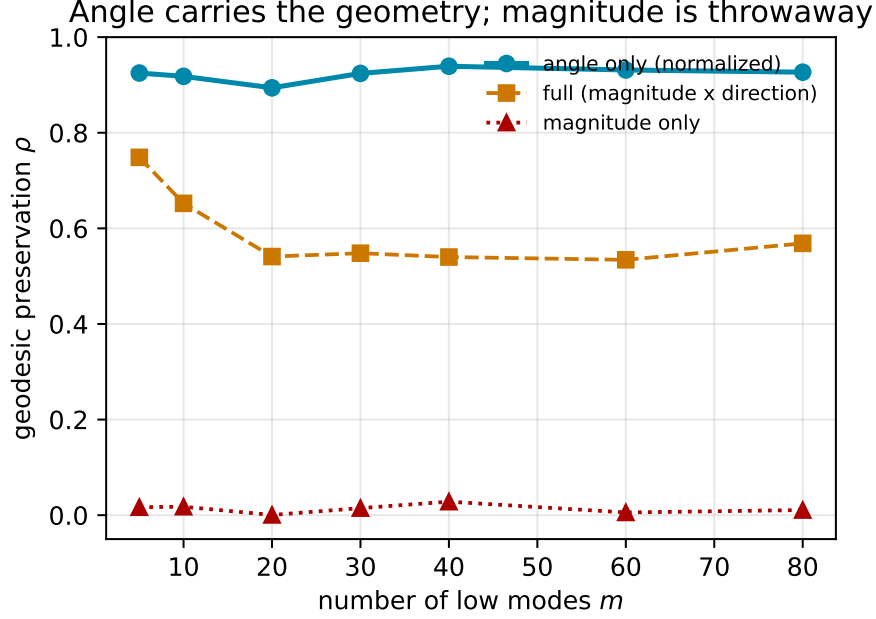


Figure 1: Spearman correlation with graph geodesics on a 2-torus ($n = 2000$) as a function of mode-count m . Angle carries the geometry; magnitude is throwaway; the full commute embedding decays with m .

5.2 Emergent 2D geometry

Random arity-2 rules yield clean but only $\approx 1.5D$ manifolds (82 manifold-grade rules in a 20k sweep; none at $d \geq 2.5$). Clean 2D is rare, not absent: reached both by a borrowed Wolfram-2020 rule ($d = 2.07 \pm 0.01$, spread 0.05, angle- $\rho = 0.82$) and by from-scratch evolutionary search (≈ 3 generations). The angular basis holds on genuine 2D emergent space.

5.3 Robustness across substrates

substrate	meas. d	angle- ρ	random	Isomap $_{d=2}$
2-torus (reference)	2	0.89	0.01	0.67
king-lattice	2	0.82	0.01	0.95
swiss-roll (embedding)	2	0.93	0.01	0.99
emergent R_{3D} (Wolfram)	2.07	0.82	0.01	0.81
causal set (Lorentzian)	3.9	0.57	0.01	0.55
small-world (destroyed)	3.6–4.3*	0.45	0.00	—

Substrates with genuine low- d Riemannian geometry all show the effect, independent of graph origin; the metric degrades where geometry is Lorentzian ($\rho = 0.57$, a degradation not a failure) or destroyed. *The small-world substrate is a heavily rewired 2D king-lattice; the rewiring inflates and scatters the three dimension estimators (ball 3.7, spectral 3.6, effective-rank 4.3; spread 0.77 versus 0.14–0.51 across the clean substrates), so no single manifold dimension is well-defined—we tabulate their range. By our own manifold-grade criterion this graph is *not* a manifold, and the low angle- ρ is the predicted failure, not an exception. The two-factor screen below makes this quantitative.

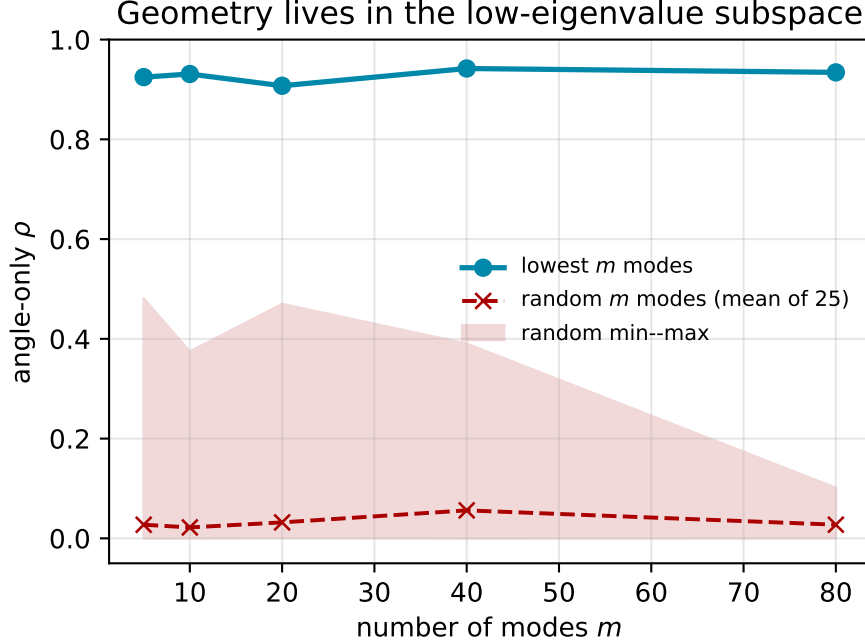


Figure 2: Geometry lives in the low-eigenvalue subspace: lowest- m modes preserve geodesics; a random-mode basis of equal size does not. The random curve is the mean over 25 draws and the shaded band its full min-max envelope—even the best-case random draw stays far below the low-mode curve.

Two caveats keep the robustness claim honest. First, on the two flat substrates (king-lattice, swiss-roll) classical MDS *beats* the angular basis (0.95 and 0.99 vs. 0.82 and 0.93): MDS wins whenever the manifold embeds near-isometrically in low-dimensional Euclidean space, while the angle wins where topology obstructs such an embedding (the torus: 0.89 vs. 0.67) and, unlike MDS, is uniform in m and n (§5.5). That 0.67 is Isomap forced into $\text{dim}=2$; matched to $\text{dim}=m$ it reaches ≈ 0.97 and edges the angle even on the torus—but Isomap is *given* the geodesic matrix as input, whereas the angular basis recovers it from the spectrum alone, so the two solve different problems. Robustness here means the angular basis never *loses* the geometry on a genuine low- d Riemannian substrate—not that it dominates every baseline on every substrate; we accordingly avoid “universal” outside the discussion. Second, the swiss-roll is a manifold-embedding case (a smooth 2D manifold carried into $\mathbb{R}^{12}$): the angular basis is the same whether the manifold comes from hypergraph rewriting or from points sampled on a continuous manifold. No sequential or temporal structure enters the graph construction, so we make no sequential-data claim.

A two-factor small-world screen. To make “destroyed” precise rather than a label, we screen every substrate on two graph statistics against degree-matched baselines—average clustering C (versus an Erdős-Rényi C_{rand} and a ring-lattice C_{latt}) and characteristic path length L (versus L_{rand})—combined into the Telesford index $\omega = L_{\text{rand}}/L - C/C_{\text{latt}}$ ($\omega < 0$ lattice-like, $\omega \approx 0$ small-world, $\omega \rightarrow 1$ random). Clean low- d manifolds sit firmly on the lattice side; the tangles do not:

substrate	angle- ρ	ω	dim-spread	verdict
2-torus	0.91	−0.55	0.14	clean
king-lattice	0.83	−0.50	0.51	clean
swiss-roll	0.93	−0.66	0.45	clean
king-lat. rewired	0.42	+0.23	0.77	small-world
causal set	0.57	+0.75	7.87	non-manifold

The screen robustly separates the genuine manifolds ($\omega \leq -0.50$) from strongly destroyed geometry ($\omega \geq +0.23$), and the inter-estimator dimension spread corroborates (≤ 0.51 clean versus ≥ 0.77 contaminated): the low angle- ρ on the rewired and Lorentzian substrates is small-world/non-manifold contamination, not a failure of the basis on genuine geometry. A king-lattice rewiring sweep both sharpens and honestly bounds this. As the rewire probability rises, angle- ρ collapses almost at once ($0.81 \rightarrow 0.47$ by 1% rewiring) while ω does not cross zero until $\approx 5\%$; so $\omega < 0$ is a *conservative* flag for destruction, and in the transition band (1–3%) angular fidelity is in fact the *more* sensitive detector of geometric corruption than either clustering or dimension spread. We therefore read the screen as confirming gross contamination, with angle- ρ itself the finer instrument—the mechanism behind the scaling-law outlier below.

5.4 The scaling law

Across 20 manifolds (5 rules $\times$ 4 seeds; the four seeds within a rule are not independent draws), $d \approx 1\text{--}3.6$,

$$\text{angle-}\rho \approx 1.09 - 0.157 d, \quad \text{slope } -0.157 \text{ (rule-clustered SE 0.061), } R^2 = 0.64, \quad d \in [1.0, 3.6].$$

(Spearman -0.881 between ρ and d is a distribution-free monotonicity check. Because the 20 points are clustered by rule, we bootstrap the slope at the *rule* level—resampling the five rules with replacement, then the four seeds within each drawn rule (with only five clusters the bootstrap is itself coarse; a caveat, not a fix)—giving an honest 95% CI of $[-0.343, -0.094]$, still excluding zero but four times wider than the naive point bootstrap $[-0.185, -0.132]$ that wrongly treats the 20 points as exchangeable. The modest $R^2 = 0.64$ reflects a single outlier—a $d \approx 1.97$ manifold at $\rho = 0.39$ —which the small-world screen of §5.3 flags as shortcut-contaminated rather than clean geometry. And because the abscissa d is itself measured (estimator spread 0.14–0.51), ordinary least squares *attenuates* the slope; an errors-in-variables correction steepens it modestly (to ≈ -0.17 under a spread-implied error variance), so -0.157 is a conservative magnitude, not an inflated one.) The fit is *empirical and local*: ρ is a rank correlation bounded by 1, so the linear form holds only within the measured range—the intercept 1.09 is an extrapolation, not a literal $d \rightarrow 0$ fidelity, and the slope may vary with substrate roughness. A bounded model fits essentially as well and respects $0 < \rho < 1$: the logistic $\rho \approx [1 + e^{0.82(d-3.66)}]^{-1}$ gives $\rho(0) = 0.95$ and $\rho \rightarrow 0$ at $R^2 = 0.62$, against the linear 0.64. Exponential forms do not help—an unoffset Ae^{-kd} still has $A = 1.18 > 1$ and an offset version drives the asymptote negative. Within range the two are visually indistinguishable; we keep the linear coefficient as the local slope and note the logistic as the boundary-respecting global form. Within range, fidelity is highest at low dimension and softens toward ≈ 0.56 at $d \approx 3.4$ (Fig. 3); the dimension-gated degeneracy of §3 is the mechanism—and Proposition 11 supplies a second, rank-specific one: the distance-ratio concentration of a d -dimensional substrate grows with d ($\mu(\kappa) = 1 - \kappa^{-d}$ for the small-ball law), so any fixed metric distortion certifies exponentially less rank structure as d rises. The worst case decays exponentially; the measured decay is linear—the eigenmap angle sits far above its distortion-only floor.

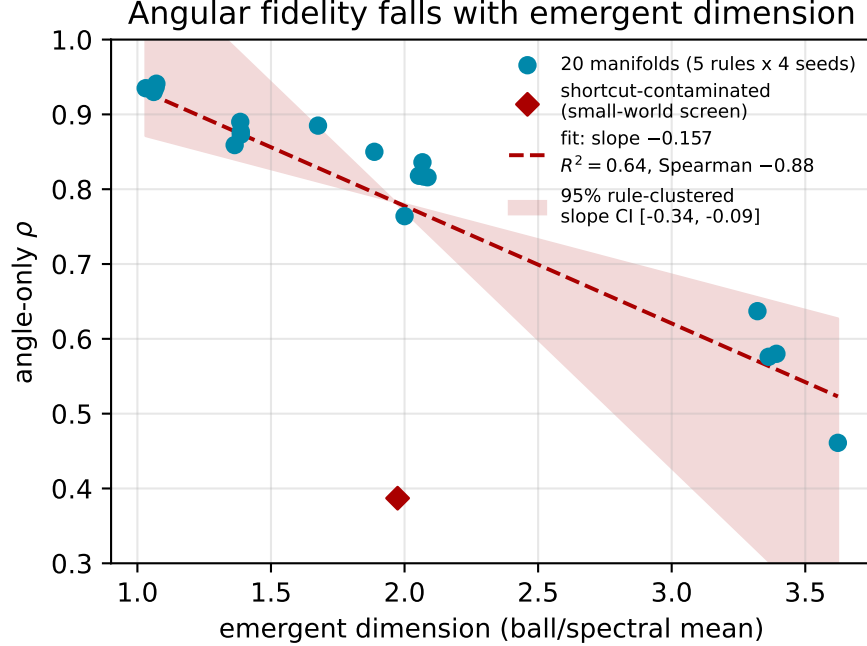


Figure 3: Angular fidelity falls linearly with emergent dimension across the 20 library manifolds (five Wolfram-model rules $\times$ four seeds, $d \approx 1\text{--}3.6$); the shaded band is the 95% *rule-clustered* slope CI $[-0.34, -0.09]$, drawn as its envelope about the data centroid—four times wider than a naive point bootstrap, since the independent unit is the rule, not the point. Per-point pair-resampling CIs are smaller than the plot markers ($\leq \pm 0.005$). The red diamond ($d \approx 1.97$, $\rho = 0.39$) is the single shortcut-contaminated manifold flagged by the small-world screen (§5.3), which drives the modest R^2 .

5.5 Verifying Corollary 1 and Conjecture 3

We check the two claims directly. The angle- ρ is flat in graph size while the full commute distance decays, and the truncated radius tracks degree ($\rho(\|\Psi_i\|, 1/\sqrt{k_i}) = 0.79\text{--}0.87$ across $m = 5\text{--}40$; radius-only $\rho \leq 0.08$). For the full *unnormalized* commute radius the degeneracy is exact, Corollary 1 holding cleanly at $d = 3$ ($\rho(R, 1/k_i + 1/k_j) \rightarrow 0.98$); $d = 2$ is borderline (log corrections); the quasi-1D case correctly shows no degeneracy, as the dimension gating predicts. Conjecture 3: in an m -sweep at $n = 4000$, angle- ρ is flat ≈ 0.93 across $m = 5\text{--}80$ while the full embedding decays $0.773 \rightarrow 0.437$.

5.6 Which weighting carries the angle?

The direction depends on the per-mode weight, so we ablate it on the torus ($d=2$) and a 3-torus ($d=3$). Plain eigenmaps ($w_k=1$) give an angle that *collapses* with mode count—angle- ρ falls from 0.55 at $m=10$ to 0.06 at $m=20$ ($d=2$)—because equal weighting lets high modes swamp the direction. The commute weight $1/\sqrt{\lambda_k}$ is flat instead ($0.91 \rightarrow 0.89$), as is diffusion $e^{-\lambda_k t}$; both suppress high modes, which is what the geometry needs. The “flat in m ” property is thus a property of the weighting, not of angles in general—and it selects the commute (or diffusion) weight, not the plain eigenmap. Proposition 14 now supplies the continuum mechanism: plain angular distances concentrate at $\sqrt{2}$ with only an oscillatory, geometry-blind correction, while the commute correction

is the (low-pass, Green-type) kernel that carries the ranks. Degree correction, by contrast, is *invisible* to the angle: dividing or multiplying each row by $\sqrt{k_i}$ (the diffusion-map convention and its opposite) leaves angle- ρ unchanged to three digits (0.889 either way), because a per-node radial factor cancels under row-normalization—a direct confirmation that the degree lives in the radius (Corollary 1), not the angle.

5.7 A bake-off against named spectral distances

The ablation isolates the *weight*; a referee rightly asks how the angle fares against the standard *distances* built from the same spectrum. We score each on the identical graph and anchor set (Spearman ρ vs. unweighted geodesics), on a 2-torus ($n=2000$) and a 3-torus ($n=3000$).

distance	2-torus	3-torus
angle (ours)	0.888	0.830
full commute / resistance	0.601	0.360
biharmonic [33]	0.735	0.515
diffusion distance (best t) [5]	0.771	0.631
amplified commute [32]	0.628	0.364
<i>Isomap, given the geodesics</i>	<i>0.971</i>	<i>0.952</i>

The angle beats every purely spectral competitor, and the margin widens in $d=3$ where commute degeneracy bites harder. The most informative row is the amplified commute distance: von Luxburg, Radl and Hein designed it precisely to undo commute degeneracy by subtracting the $1/k_i+1/k_j$ term, yet it barely improves on raw commute (0.628 vs. 0.601) and stays far below the angle. *Subtracting* the degenerate radial term is not the same as *dividing it out*: row-normalization removes the whole degree-dependent radial factor at once, which is what the geometry needs (Lemma 4, Corollary 1). Isomap wins, but only because it is handed the geodesics as input—it is an oracle ceiling, not a competitor; the angle recovers 0.83–0.89 of that ceiling from the graph alone. We do not tabulate the p -resistance family [35] as a separate row because its informative endpoints are already present: $p=2$ is the full-commute row, and $p=1$ is the shortest-path distance—i.e. the geodesic *target* itself, not a competitor. The bake-off’s scope is closed-form spectral distances; intermediate p has no closed form (each pair needs its own convex q -Dirichlet solve), so it buys no comparison the two endpoints do not already frame.

5.8 Radius is density, angle is geometry: a non-uniform torus

The decomposition “degree/density in the radius, geometry in the angle” predicts a sharp behaviour under *non-uniform* sampling that uniform substrates cannot test. We sample the flat 2-torus with density $p(x) \propto 1 + a \cos 2\pi x$, build a fixed-radius geometric graph, and sweep the contrast a (degree ratio up to $45\times$ at $a=0.9$), scoring the angle against the *true* flat-torus geodesics—the intrinsic metric, which is unchanged by how we sample it. As predicted, the radius tracks the imposed density profile ($\rho(\|\Psi\|, 1/\sqrt{p}) = 0.53\text{--}0.67$), confirming it is a density coordinate. The raw angle, however, is *not* sampling-invariant: angle- ρ to the true metric degrades from 0.93 to 0.79 as the contrast grows, because the graph Laplacian converges to a density-weighted operator, not the Laplace–Beltrami operator [5, 34]. The classical fix recovers it exactly: the Coifman–Lafon $\alpha=1$ normalization $W \mapsto D^{-1}WD^{-1}$ (with D the density estimate) removes the density weight, and the density-normalized angle is restored to 0.92–0.96, now *flat* across the whole contrast range. So the angle carries the geometry, but only once the density is properly quotiented out—and the same normalization that fixes the angle is the one that isolates the density into the radius.

5.9 Stability across substrate evolution

Robustness across substrate *types* (§5) is a statement about fixed graphs. The angular basis also persists as a substrate *evolves*. For each of 151 qualifying Wolfram-model rules (95 arity-2, 56 arity-3) we embed successive rewrite snapshots on a fixed anchor set and compare, between consecutive snapshots, the Spearman persistence of the *angular* pairwise structure (**consec**) against the persistence of the underlying *geodesic* structure itself (**churn**)—the substrate’s own rate of change is the baseline the basis must keep pace with. On all 143 rules resolvable at protocol scale (8, all arity-3, never yield ≥ 3 distinct snapshots and are excluded from the denominator but reported; all 95 arity-2 rules resolve), the angular structure tracks the evolving geometry at least as stably as the graph’s own geodesics, $\overline{\text{consec}} \geq 0.9 \overline{\text{churn}}$: a 143/143 = 100% pass against a pre-registered 80% bar, with zero substrate-unstable rules. Over the longest time-lever (first vs. last snapshot) angular persistence declines only gently with emergent dimension,

$$\text{slope} = -0.029/\text{dim}, \quad \text{Spearman} = -0.48, \quad d \in [0.98, 2.23],$$

and this replicates independently in both arities (arity-2 $-0.028/\text{dim}$, $\rho = -0.45$, $n = 95$; arity-3 $-0.029/\text{dim}$, $\rho = -0.54$, $n = 48$)—the same mechanism, unchanged by moving from binary to ternary rewriting.²

5.10 Continuum survival, and a dissociation

Every result so far is at finite N ($\approx 2\text{--}4\text{k}$ nodes, capped by dense **eigh**); the “uniform in N ” clause of Corollary 1 and Conjecture 3 is untested at scale (§5.5). We test it directly by refining each substrate toward the continuum with a sparse eigensolver, $N \in \{2, 5, 10, 20, 50\} \times 10^3$, four seeds, $m = 10$ modes, $n = 300$ anchors—well past the $\sim 5\text{k}$ dense ceiling. The two substrate families dissociate. On genuine Riemannian substrates the basis *sharpens*: on random geometric graphs the angular fidelity is flat-to-rising in N , most cleanly in the higher-dimensional case,

$$\begin{aligned} \text{RGG } d=3: \quad \text{angle-}\rho: 0.856 \rightarrow 0.912 \quad (N: 2\text{k} \rightarrow 50\text{k}), \\ \text{Spearman}(\text{angle-}\rho, \log N) = +1.00, \end{aligned}$$

with $d=2$ likewise monotone ($+0.90, 0.916 \rightarrow 0.929$): refining the mesh makes the low modes resolve the geometry *better*, not worse, and both controls stay pinned (higher-mode $\rho \leq 0.05$, magnitude $\rho \leq 0.03$, eigensolver residual $\sim 10^{-14}$) at every N . The finite- N evidence of §§5–5.5 was not a small-graph artifact.

The dissociation is with *emergent* substrates. The $\text{R}_{3\text{D}}$ Wolfram-model manifold—geometry with no latent coordinates—*decays* monotonically with refinement, angle- $\rho: 0.749 \rightarrow 0.654$ (Spearman = -1.00), falling below a pre-registered floor of 0.75; and at the largest N its magnitude control begins to break the clean angle/radius separation that holds exactly on the RGGs (one 50k seed crosses $\rho_{\text{mag}} = 0.33$ and is gated out; the sub-floor decay verdict holds with or without it). A sampled manifold refines to a clean continuum; an emergent hypergraph, under this basis, does not—its fine structure drifts away from a manifold as it grows. We read this not as a limitation but as a result: the angular basis is a *diagnostic* for whether a substrate’s fine structure stays manifold-like under refinement, and it returns different verdicts for sampled (2/3 pass) and emergent (1/1 fail) geometry.

²Pre-registered as Amendment 2: an earlier capture hook could append the same state once per crossed edge-count threshold, inflating consecutive stability toward a spurious 1.0. The corrected rule takes at most one snapshot per generation with dynamic $2\times$ spacing; the numbers here are from that harness.

One qualification is essential, and it deepens rather than weakens the reading: the dissociation is stated at a *fixed* observer budget ($m = 10$ while N grows $5\times$). Rerunning with m *spectral-gap-matched*—holding the resolved eigenvalue scale τ fixed, so m grows with N (to $m \approx 47$ at $N=10^4$)—inverts it: the RGG now *falls* ($0.856 \rightarrow 0.775$, Spearman = -1.0), because the added high modes inject sub-geodesic detail, while the emergent substrate stops decaying ($0.747 \rightarrow 0.692$, monotonicity gone, Spearman = -0.5). No single budget is optimal for both. So the dissociation is a property of the *fixed-budget* observer, not an absolute substrate invariant: a manifold keeps its geodesic geometry in a fixed low-mode band (refining sharpens those modes), whereas the emergent substrate spreads geometry across scales (a fixed observer resolves an ever-smaller fraction as it grows). “Clean geometry” is thus a relation between observer resolution and substrate scale—the sharpest form of the mode-budget consistency the angular basis exhibits (§5.5), and the motivation for a dimension-adaptive basis (§7). Modest evidence—one emergent substrate, one seed, $N \leq 10^4$ —so we state the direction, not a law.³

6 Honest Negatives

Not secretly hyperbolic (rejected). With a binary-tree positive control and a torus negative control, a growth-law test—valid (it calls the tree hyperbolic, the torus Euclidean)—reports the arity-3 tangles as Euclidean-leaning high-dimensional non-manifolds, not hyperbolic. **Curvature is not dynamical (null).** Ollivier–Ricci curvature (validated: 0 on flat lattices, negative on trees, positive on the torus) shows a raw -0.98 against update density that deflates to a tautology (update-count equals degree, $\rho = +1.00$; hubs are negatively curved); controlling for degree, the independent signal is partial- $\rho = -0.14$. This *bounds the observer-theory interpretation*—no emergent Einstein tensor—and leaves the spectral result untouched.

7 Discussion and Conclusion

The primary result is basis-level and substrate-agnostic. The angular coordinate of the low-Laplacian embedding preserves geodesics while the radial coordinate is von-Luxburg-degenerate noise. This gives Ng–Jordan–Weiss row-normalization a geodesic justification complementary to the cluster-separation theory of [14], places it in the same scale-from-direction decomposition that underlies vector and KV-cache compression [12, 10] (with the scope boundary of §1: the quotient, not the discard, is what recurs), and holds across every substrate family we tested with genuine low- d Riemannian geometry, independent of graph origin. That the same basis works identically on hypergraph-rewriting and manifold-sampled substrates is evidence that scale-invariant angular coarse-graining is a general mechanism by which relational data carries geometric structure.

A boundary from compression. The present result is not a general theorem that angular coordinates always carry the semantics. In KV-cache compression, per-vector angular quantization of attention *keys* can score well under reconstruction metrics and still destroy generation, because the attention geometry $QK^\top$ depends on per-channel scale; per-channel treatment of the keys recovers quality [10]. The consumer-relative synthesis [11] names and demonstrates the mechanism: discarding a direction the consumer reads is an *omission*, and an omission incurs an irreducible

³Pre-registered (PREREG_RUNG4) before the run; survival bar $\text{angle-}\rho(N_{\max}) \geq \max(0.75, 0.9 \text{angle-}\rho(N_{\min}))$ on $\geq 2/3$ substrates. Amendment 1, registered before any run: the whole-spectrum random-mode control needs a dense decomposition unavailable at $N \geq 50k$, so it is the top m of the lowest-64 sparse-computed modes—a faithful, sparse-affordable stand-in, same ≤ 0.15 bar.

downstream *distortion floor* that no bit budget removes — shown on a trained Llama attention layer (softmax divergence $800\text{--}5700\times$ that of a rate-matched consumer-aware coder), the operational form of a violated (A2). This is not a counterexample to our principle but its sharpest confirmation: row-normalization is justified precisely when the discarded radial variable is nuisance structure for the metric being preserved—condition (A2) of Theorem 6, stated for the downstream metric. For graph geodesics the radius is degree/density noise and the quotient is correct; for embedding retrieval the norm is worth storing alongside the direction; for attention keys the per-channel scales *are* the geometry. Keep the geometry: sometimes that is the angle, sometimes angle plus norm, sometimes per-channel scale.

One interpretation is observer-theoretic. In Wolfram’s observer-relative framework, the coarse-graining a bounded observer performs to perceive geometry is exactly this angular projection—making “the observer’s basis” a concrete candidate for the coarse-graining the program leaves unspecified. We offer this as motivation and bound it honestly: its strongest claim—dynamical, energy- responsive curvature—is a null (§6). The core contribution stands independent of the physics reading. Open problems: proving Conjecture 3 (in its rank form, and ideally the bi-Lipschitz strengthening—both now reduced at fixed m to the chart hypotheses (H1)–(H3) plus, for rank, a distance-ratio condition (Corollary 9, Proposition 11); the open core is now the *scale-dependent* commute uniformity—fixed-scale uniformity in m is proven for the heat filter and refuted for the commute filter (Theorem 12, Proposition 13)—graph transfer at growing spectral windows (Appendix C), and verifying (H1)–(H3) beyond the flat torus, where Corollary 10 settles them); extending Corollary 1 to the truncated, normalized radius actually used in practice; synthesizing clean manifolds above $d \approx 2$; the exploratory probe’s deferred controls—a generic-text baseline at matched n and model, and k -sensitivity (Appendix D); and a dimension-adaptive angular basis that tunes mode-count and diffusion time to counter the fidelity decay. Keep the angle, drop the magnitude—*when the magnitude is nuisance for the metric being preserved*, which for graph geodesics it provably is: the basis-level fact is the contribution; the observer interpretation is where we leave the door open.

Reproducibility. The pre-registration documents (PREREG_RUNG3/4 with dated amendments), all experiment scripts and per-run result files (the torus and substrate benchmarks, the weighting ablation, the small-world screen, the continuum and scaled-budget refinements, the truncation/normalization checks, and the cross-domain probe), the manifold-rule sets behind the 20-point scaling law (five rules—R_2D, R_frac, evolved-2D, R_3D, R_SR—each at four seeds) and the 151 evolution rules, and the random seeds and solver settings are released in the accompanying repository, <https://github.com/ahb-sjsu/the-angular-observer>. We state the reproducibility tiers plainly. *Fully reproducible locally* (NumPy/SciPy/NetworkX, CPU): the torus core result and controls, the substrate benchmarks and two-factor screen, the weighting ablation, the spectral-distance bake-off, the non-uniform-density torus, the truncation/normalization checks, and the clustered-bootstrap scaling-law analysis—each script emits the exact committed result JSON. *Requires a cluster*: the $N \leq 5 \times 10^4$ continuum refinements (§5.10). *Not yet externally reproducible*: the scaled-budget refinement’s builders and the cross-domain probe’s embeddings (pending a vendored export)—which is why the probe is marked exploratory and kept out of the main evidence chain (Appendix D).

A Proof of Corollary 1

A referee correctly observed that the diagonal statement $L_{ii}^+ \rightarrow 1/k_i$ does not follow from the pairwise resistance limits alone: $R(i, j) = L_{ii}^+ + L_{jj}^+ - 2L_{ij}^+$ determines only those combinations, and a summation argument must control the error terms it accumulates. We give the argument in full. Two inputs are used, both explicit in the corollary's statement: **(i)** the *uniform multiplicative* form of the von Luxburg–Radl–Hein degeneracy—for $d \geq 3$ graphs in their model there is a sequence $\eta_n \rightarrow 0$ with

$$R(i, j) = \left(\frac{1}{k_i} + \frac{1}{k_j} \right) (1 + \eta_{ij}), \quad \sup_{i \neq j} |\eta_{ij}| \leq \eta_n$$

(this is the form in which [16] state their result, with rates); and **(ii)** a bounded degree ratio, $k_{\max}/k_{\min} \leq K$ for all n (their graph models satisfy this with high probability). Write $S := \sum_j 1/k_j$ and note $S/n \leq 1/k_{\min} \leq K/k_i$ for every i by (ii).

Step 1 (row sums). L^+ annihilates constants ($L\mathbf{1} = 0$ and L^+ shares the range of L on $\mathbf{1}^\perp$), so $\sum_j L_{ij}^+ = 0$: the gauge. Summing $R(i, j)$ over $j \neq i$ and using the gauge,

$$\sum_{j \neq i} R(i, j) = (n-1)L_{ii}^+ + (\text{tr } L^+ - L_{ii}^+) - 2 \sum_{j \neq i} L_{ij}^+ = n L_{ii}^+ + \text{tr } L^+,$$

since $\sum_{j \neq i} L_{ij}^+ = -L_{ii}^+$. By (i) the same sum equals

$$\sum_{j \neq i} \left(\frac{1}{k_i} + \frac{1}{k_j} \right) (1 + \eta_{ij}) = \frac{n-2}{k_i} + S + e_i, \quad |e_i| \leq \eta_n \left(\frac{n}{k_i} + S \right).$$

Equating and dividing by n ,

$$L_{ii}^+ = \frac{1}{k_i} + C_n - \frac{2}{nk_i} + \frac{e_i}{n}, \quad C_n := \frac{S - \text{tr } L^+}{n} \quad (\text{independent of } i). \quad (\text{A.1})$$

Step 2 (trace identity kills C_n). Summing (A.1) over i : $\text{tr } L^+ = S + nC_n - \frac{2}{n}S + \frac{1}{n} \sum_i e_i$. Substituting $nC_n = S - \text{tr } L^+$ and solving,

$$\text{tr } L^+ = S \left(1 - \frac{1}{n} \right) + \frac{1}{2n} \sum_i e_i, \quad \text{hence} \quad C_n = \frac{S}{n^2} - \frac{1}{2n^2} \sum_i e_i.$$

Since $|\sum_i e_i| \leq \eta_n \sum_i (n/k_i + S) = 2\eta_n nS$, we get $|C_n| \leq S/n^2 + \eta_n S/n \leq (1/n + \eta_n) K/k_i$ for every i , using (ii).

Step 3 (conclusion). Feeding the bounds on C_n and e_i/n back into (A.1),

$$\left| L_{ii}^+ - \frac{1}{k_i} \right| \leq \frac{(1/n + \eta_n)K}{k_i} + \frac{2}{nk_i} + \frac{\eta_n}{k_i} (1 + K) \leq \frac{C(K)(\eta_n + 1/n)}{k_i},$$

so $k_i L_{ii}^+ \rightarrow 1$ uniformly over nodes, i.e. $L_{ii}^+ = (1/k_i)(1 + o(1))$ and $\|\Phi_i\| = \sqrt{L_{ii}^+} = (1/\sqrt{k_i})(1 + o(1))$. $\square$

Without (ii) the conclusion weakens gracefully rather than failing: the same computation gives $L_{ii}^+ = 1/k_i + O((\eta_n + 1/n)S/n)$, an *additive* error at the scale of the harmonic mean degree, so the degeneracy statement survives at the level of pairwise resistance ($R \approx 1/k_i + 1/k_j$, which is (i) itself) even when per-node relative control is lost. The $j=i$ diagonal term, where $R(i, i) = 0$ while $1/k_i + 1/k_j = 2/k_i$, is excluded from the sums above, which is why the $(n-2)/k_i$ bookkeeping appears.

B Proof of Theorem 8

Setting and notation

M is a compact d -dimensional Riemannian manifold without boundary, volume normalized to 1; $(\mu_k, \varphi_k)_{k \geq 0}$ are the eigenpairs of its (positive) Laplace–Beltrami operator, $0 = \mu_0 < \mu_1 \leq \mu_2 \leq \dots$, with $\{\varphi_k\}$ orthonormal in $L^2(M)$. Under non-uniform sampling with an admissible density the identical argument runs against the density-weighted operator that the graph Laplacian actually converges to [3, 23] (the $\alpha=1$ normalization of §5.8 restores the geometric operator), with (H1)–(H2) read for its eigenpairs. The continuum objects are

$$F(x) = \left(\frac{\varphi_k(x)}{\sqrt{\mu_k}} \right)_{k=1}^m, \quad R(x) = \|F(x)\|, \quad N(x) = F(x)/R(x),$$

and m sits at a spectral gap: $\mu_m < \mu_{m+1}$. Write $\Phi_m := \max_{k \leq m} \|\varphi_k\|_\infty < \infty$ (eigenfunctions on a smooth compact manifold are smooth). The sample is $x_1, \dots, x_n$ i.i.d. from the stated density class, G_n the ε -graph (or k -NN graph) on it, and (λ_k, v_k) the eigenpairs of L_{sym} with the empirical normalization $\frac{1}{n} \sum_i v_k(i)^2 = 1$. The graph objects are $\Psi_i = (v_k(i)/\sqrt{\lambda_k})_{k=1}^m$ and $\hat{u}_i = \Psi_i/\|\Psi_i\|$.

Step 0: three invariances

The claim concerns only the pairwise distances $\|\hat{u}_i - \hat{u}_j\|$. These are invariant under (i) a global rescaling of all eigenvectors (the choice of normalization), (ii) a global rescaling of all eigenvalues (which rescales every weight $1/\sqrt{\lambda_k}$ by the same factor), and (iii) a fixed orthogonal transformation $Q \in O(m)$ applied to every Ψ_i (then $\|Q\hat{u}_i - Q\hat{u}_j\| = \|\hat{u}_i - \hat{u}_j\|$). We use (i) to work in the empirical normalization, (ii) to apply the deterministic graph-to-continuum eigenvalue scaling c_n of [23], and (iii) to absorb the basis alignment of Lemma 17.

Step 1: the continuum angular map is locally bi-Lipschitz

Lemma 15. *Under (H1)–(H2), for all $x, y \in M$ with $d_M(x, y) \leq r_0$,*

$$a d_M(x, y) \leq \|N(x) - N(y)\| \leq b d_M(x, y), \quad a := \frac{\sqrt{L_0^{-2} - \Lambda_0^2}}{\rho^+}, \quad b := \frac{L_0}{\rho^-}.$$

Proof. Apply the identity of Lemma 4 to $u = F(x)$, $v = F(y)$. By (H1), $L_0^{-2} d_M^2 \leq \|F(x) - F(y)\|^2 \leq L_0^2 d_M^2$; by (H2), $(R(x) - R(y))^2 \leq \Lambda_0^2 d_M^2$ and $(\rho^-)^2 \leq R(x)R(y) \leq (\rho^+)^2$. The ratio in the identity is therefore at least $(L_0^{-2} - \Lambda_0^2) d_M^2 / (\rho^+)^2$ —positive precisely because $\Lambda_0 < 1/L_0$ —and at most $L_0^2 d_M^2 / (\rho^-)^2$. $\square$

Note $a \leq b$: a two-sided constant satisfies $L_0^{-1} \leq L_0$, so $L_0 \geq 1$, hence $\sqrt{L_0^{-2} - \Lambda_0^2} \leq L_0^{-1} \leq L_0$, and $\rho^- \leq \rho^+$.

Step 2: normalization is stable

Lemma 16. *For nonzero $u, w \in \mathbb{R}^m$,*

$$\left\| \frac{u}{\|u\|} - \frac{w}{\|w\|} \right\| \leq \frac{2\|u - w\|}{\max(\|u\|, \|w\|)}.$$

Proof. $\frac{u}{\|u\|} - \frac{w}{\|w\|} = \frac{u-w}{\|u\|} + w \left(\frac{1}{\|u\|} - \frac{1}{\|w\|} \right)$, and the second term has norm $|\|w\| - \|u\||/\|u\| \leq \|u - w\|/\|u\|$. So the left side is at most $2\|u - w\|/\|u\|$; by symmetry in $u \leftrightarrow w$, also at most $2\|u - w\|/\|w\|$. $\square$

Step 3: sup-norm spectral transfer

This is the step for which L^2 consistency [3] does not suffice and the ∞ -norm rates of [23] are the right tool. Group $\mu_1 \leq \dots \leq \mu_m$ into blocks of equal eigenvalues (multiplets); because m sits at a spectral gap, the last block ends exactly at m .

Lemma 17 (Calibration). *Under the sampling model of [23] there are sequences $\epsilon_n, \delta_n \rightarrow 0$ (the eigenvector- ∞ -norm and eigenvalue rates of that paper, the former carrying a factor $\sqrt{m}$) and events $\mathcal{A}_n$ with $\mathbb{P}(\mathcal{A}_n) \rightarrow 1$ on which, after the eigenvalue scaling $\tilde{\lambda}_k := \lambda_k/c_n$ of invariance (ii), there is an orthogonal $Q_n \in O(m)$, block-diagonal across the multiplets, with*

$$\max_{k \leq m} |\tilde{\lambda}_k - \mu_k| \leq \delta_n, \quad \max_{i \leq n} \|Q_n v(i) - \varphi(x_i)\| \leq \epsilon_n,$$

where $v(i) = (v_1(i), \dots, v_m(i))$ and $\varphi(x) = (\varphi_1(x), \dots, \varphi_m(x))$. Consequently, with $\tilde{\Psi}_i := (v_k(i)/\sqrt{\tilde{\lambda}_k})_{k=1}^m$ (the same angle as Ψ_i , by invariances (i)–(ii)) and provided $\delta_n \leq \mu_1/2$,

$$\max_{i \leq n} \|Q_n \tilde{\Psi}_i - F(x_i)\| \leq E_n := \frac{\epsilon_n}{\sqrt{\mu_1}} + \frac{2\delta_n}{\mu_1^{3/2}} \sqrt{m} (\Phi_m + \epsilon_n) \rightarrow 0.$$

In particular the graph radius inherits the continuum band: $\rho^- - E_n \leq \|\tilde{\Psi}_i\| \leq \rho^+ + E_n$ for every node i .

Proof. (a) *Spectral consistency.* For the unnormalized and random-walk graph Laplacians, [23] prove, with probability tending to one, eigenvalue convergence $|\tilde{\lambda}_k - \mu_k| \leq \delta_n$ and eigenvector consistency in ∞ -norm with explicit rates, stated per eigenvalue cluster: for each multiplet there is an orthogonal alignment of the discrete eigenvectors of that cluster with an $L^2(M)$ -orthonormal basis of the limiting eigenspace such that each aligned discrete eigenvector is $(\epsilon_n/\sqrt{m})$ -close in sup-norm to its continuum partner. (Multiplicity is exactly why the alignment is block-orthogonal rather than a per-mode sign choice; a simple eigenvalue is the 1×1 block, where it *is* a sign choice.) Collecting the m per-mode bounds into the vector $v(i)$ gives $\|Q_n v(i) - \varphi(x_i)\| \leq \epsilon_n$.

(b) *From the random-walk to the symmetric normalization.* $L_{\text{rw}} = D^{-1}(D - A) = D^{-1/2} L_{\text{sym}} D^{1/2}$; the two matrices are similar, so their *eigenvalues coincide exactly* and only the eigenvectors differ, by $v_k^{\text{sym}} = D^{1/2} v_k^{\text{rw}}$ up to normalization. Under the stated density conditions the degrees concentrate uniformly— $\max_i |k_i/\bar{k}_n - 1| = O(\sqrt{\log n / (n \epsilon_n^d)})$ for the uniform-density ϵ -graph by a Bernstein-plus-union bound [34, 23], the degree of a k -NN graph being k by construction up to symmetrization, and the non-uniform case holding against the weighted limit operator per the Setting paragraph—so entrywise $|v_k^{\text{sym}}(i) - v_k^{\text{rw}}(i)| \leq C \eta_n (\Phi_m + 1)$ with $\eta_n \rightarrow 0$; this perturbation is absorbed into ϵ_n .

(c) *Bookkeeping.* Let $W_\mu = \text{diag}(\mu_k^{-1/2})$ and $\tilde{W} = \text{diag}(\tilde{\lambda}_k^{-1/2})$, so $F(x) = W_\mu \varphi(x)$ and $\tilde{\Psi}_i = \tilde{W} v(i)$. Since W_μ is a scalar multiple of the identity on each multiplet and Q_n is block-diagonal across the multiplets, $Q_n W_\mu = W_\mu Q_n$. Hence

$$Q_n \tilde{\Psi}_i - F(x_i) = Q_n (\tilde{W} - W_\mu) v(i) + W_\mu (Q_n v(i) - \varphi(x_i)).$$

For the first term, $\delta_n \leq \mu_1/2$ gives $\tilde{\lambda}_k \geq \mu_1/2$, so

$$\|\tilde{W} - W_\mu\|_{\text{op}} = \max_{k \leq m} \frac{|\mu_k - \tilde{\lambda}_k|}{\sqrt{\tilde{\lambda}_k} \mu_k (\sqrt{\tilde{\lambda}_k} + \sqrt{\mu_k})} \leq \frac{2\delta_n}{\mu_1^{3/2}},$$

while $\|v(i)\| \leq \|Q_n v(i) - \varphi(x_i)\| + \|\varphi(x_i)\| \leq \epsilon_n + \sqrt{m} \Phi_m \leq \sqrt{m} (\Phi_m + \epsilon_n)$. For the second term, $\|W_\mu\|_{\text{op}} = \mu_1^{-1/2}$. Adding the two bounds gives E_n . The radius statement is the reverse triangle inequality $|\|\tilde{\Psi}_i\| - R(x_i)| = |\|Q_n \tilde{\Psi}_i\| - \|F(x_i)\|| \leq E_n$ together with $\rho^- \leq R \leq \rho^+$ from (H2). $\square$

General filters (proof of the calibration claim in Corollary 7). The proof above uses only two properties of the weight matrix $W_\mu = \text{diag}(\mu_k^{-1/2})$: it is constant on multiplets (hence commutes with the block-orthogonal Q_n), and it is Lipschitz in the eigenvalue on the retained window. Both hold for any positive Lipschitz filter h : replacing W_μ by $H := \text{diag}(h(\mu_k))$ and $\widetilde{W}$ by $\text{diag}(h(\tilde{\lambda}_k))$, step (c) goes through verbatim with $\|\text{diag}(h(\tilde{\lambda}_k)) - H\|_{\text{op}} \leq L_h \delta_n$ and $\|H\|_{\text{op}} \leq H_m$, giving $E_{n,h} = H_m \epsilon_n + L_h \delta_n \sqrt{m} (\Phi_m + \epsilon_n)$ as claimed; the commute filter $h(\lambda) = \lambda^{-1/2}$ (with $L_h \leq 2\mu_1^{-3/2}$ on $[\mu_1/2, \infty)$ and $H_m = \mu_1^{-1/2}$) recovers the E_n displayed above.

Step 4: assembly

Proof of Theorem 8. Work on the event $\mathcal{A}_n$ of Lemma 17, with n large enough that $E_n < \rho^-/2$ (so $\tilde{\Psi}_i \neq 0$ and every $\hat{u}_i$ is defined). By invariance (iii) we may compare angles through the calibrated embedding: writing $\hat{q}_i := Q_n \tilde{\Psi}_i / \|Q_n \tilde{\Psi}_i\| = Q_n \hat{u}_i$, Lemma 16 with $u = Q_n \tilde{\Psi}_i$, $w = F(x_i)$, and $\|F(x_i)\| \geq \rho^-$ gives

$$\|\hat{q}_i - N(x_i)\| \leq \frac{2E_n}{\rho^-} \quad \text{for every node } i.$$

Set $\theta_n := 8E_n/(\rho^-) \rightarrow 0$, and fix a sampled pair with $\theta_n \leq d := d_M(x_i, x_j) \leq r_0$. The pairwise angular error is at most $4E_n/\rho^- = (a/2)\theta_n \leq (a/2)d$, so by Lemma 15, the triangle inequality, and $\|\hat{u}_i - \hat{u}_j\| = \|\hat{q}_i - \hat{q}_j\|$,

$$\|\hat{u}_i - \hat{u}_j\| \geq a d - \frac{a}{2} d = \frac{a}{2} d = A d, \quad \|\hat{u}_i - \hat{u}_j\| \leq b d + \frac{a}{2} d \leq 2b d = B d,$$

the last inequality using $a \leq b$ (Step 1). Both bounds hold simultaneously for *all* such pairs because the calibration of Lemma 17 is uniform over nodes, and $\mathbb{P}(\mathcal{A}_n) \rightarrow 1$. $\square$

Step 5: from local to global (proof of Corollary 9)

Proof of Corollary 9. The map $(x, y) \mapsto \|N(x) - N(y)\|$ is continuous (F is continuous and $R \geq \rho^- > 0$) and, by (H3), strictly positive on the compact set $\{(x, y) \in M \times M : d_M(x, y) \geq r_0\}$; its minimum c_0 is therefore attained and positive. Work on the event of Lemma 17 with n large enough that $E_n < \rho^-/2$ and $4E_n/\rho^- \leq c_0/2$. For sampled pairs with $\theta_n \leq d_M \leq r_0$, Theorem 8 gives the bounds with (A, B) , hence with (A_g, B_g) . For sampled pairs with $d := d_M > r_0$: the calibration bound $\|\hat{q}_i - N(x_i)\| \leq 2E_n/\rho^-$ of Step 4 gives

$$\|\hat{u}_i - \hat{u}_j\| \geq \|N(x_i) - N(x_j)\| - \frac{4E_n}{\rho^-} \geq c_0 - \frac{c_0}{2} = \frac{c_0}{2} \geq \frac{c_0}{2 \text{diam } M} d \geq A_g d,$$

while unit vectors satisfy $\|\hat{u}_i - \hat{u}_j\| \leq 2 \leq (2/r_0) d \leq B_g d$. $\square$

Step 6: the flat torus (proof of Corollary 10)

Proof of Corollary 10. On $M = \mathbb{R}^2/(2\pi\mathbb{Z})^2$ with volume normalized to 1, the Laplace–Beltrami eigenvalues are $|k|^2$, $k \in \mathbb{Z}^2$: the first nontrivial eigenvalue $\mu = 1$ has multiplicity four, spanned by the L^2 -orthonormal functions $\sqrt{2}\cos x$, $\sqrt{2}\sin x$, $\sqrt{2}\cos y$, $\sqrt{2}\sin y$, and the next eigenvalue is 2—so $m = 4$ exhausts the first multiplet and sits at a spectral gap. The truncated eigenmap (weights $1/\sqrt{\mu_k} = 1$) is the Clifford embedding $F(x, y) = \sqrt{2}(\cos x, \sin x, \cos y, \sin y)$.

(H2). $\|F\|^2 = 2(\cos^2 x + \sin^2 x) + 2(\cos^2 y + \sin^2 y) = 4$: the radius is constant, $R \equiv 2$, so $\Lambda_0 = 0 < 1/L_0$ and $\rho^- = \rho^+ = 2$.

(H1), globally. Using $(\cos a - \cos b)^2 + (\sin a - \sin b)^2 = 4 \sin^2(\frac{a-b}{2})$,

$$\|F(p) - F(q)\|^2 = 8 \left[\sin^2 \frac{\Delta x}{2} + \sin^2 \frac{\Delta y}{2} \right], \quad \Delta x, \Delta y \in [-\pi, \pi],$$

while $d_M(p, q)^2 = \Delta x^2 + \Delta y^2$. The elementary bounds $\frac{2}{\pi}|\delta| \leq 2|\sin(\delta/2)| \leq |\delta|$ on $[-\pi, \pi]$ give

$$\frac{2\sqrt{2}}{\pi} d_M(p, q) \leq \|F(p) - F(q)\| \leq \sqrt{2} d_M(p, q) \quad \text{for all } p, q,$$

so (H1) holds with $L_0 = \sqrt{2}$ (indeed $L_0^{-1} = 1/\sqrt{2} \leq 2\sqrt{2}/\pi$) and $r_0 = \text{diam } M = \pi\sqrt{2}$.

(H3). $N = F/2$; since the radius is constant, $N(p) = N(q)$ iff $F(p) = F(q)$ iff $e^{ix_p} = e^{ix_q}$ and $e^{iy_p} = e^{iy_q}$ iff $p = q$.

Because $r_0 = \text{diam } M$, Theorem 8 alone is already global, with $A = \sqrt{L_0^{-2}}/(2\rho^+) = 1/(4\sqrt{2})$ and $B = 2L_0/\rho^- = \sqrt{2}$. The sharp continuum distortion of N is the exact ratio range above: $\|N(p) - N(q)\|/d_M(p, q) \in [\sqrt{2}/\pi, 1/\sqrt{2}]$, of width $\pi/2$, the extremes attained along a coordinate direction at antipodal ($|\delta| = \pi$) and infinitesimal separations respectively. Basis-independence: any L^2 -orthonormal basis of the $\mu = 1$ eigenspace equals Q applied to the Clifford one for some $Q \in O(4)$, and angular pairwise distances are Q -invariant (Step 0, invariance (iii)). $\square$

Step 7: the small-ball ratio law (used by Proposition 11)

Let D_1, D_2 be i.i.d. with the small-ball distance law $F(r) = (r/r_0)^d$ on $[0, r_0]$ (density $\propto r^{d-1}$, the flat approximation to pair distances within a geodesic ball). For $\kappa \geq 1$,

$$\mathbb{P}(D_2 \geq \kappa D_1) = \int_0^{r_0} F\left(\frac{y}{\kappa}\right) dF(y) = \frac{1}{\kappa^d} \int_0^{r_0} \left(\frac{y}{r_0}\right)^d \frac{dy^{d-1}}{r_0^d} dy = \frac{1}{2\kappa^d},$$

so by symmetry $\mathbb{P}(\max/\min \geq \kappa) = \kappa^{-d}$ and $\mu(\kappa) = 1 - \kappa^{-d}$ exactly. The rank certificate of Proposition 11 is thus $\tau \geq 2\kappa^{-d} - 1$, $\rho_S \geq 3\kappa^{-d} - 2$: exponential decay of the worst case in the intrinsic dimension.

Why the resolution cutoff is necessary

The cutoff θ_n is forced by the statement's generality, not by the method. In a hard-threshold ε -graph, two sample points at distance below the bandwidth can be *combinatorial twins*—identical neighbourhoods, hence identical rows of A and equal degrees. For twins i, j the difference $e_i - e_j$ is itself an eigenvector of L_{sym} —at eigenvalue 1 for non-adjacent twins and $1 + 1/k$ for adjacent ones—so every eigenvector at any *other* eigenvalue is orthogonal to it and has identical entries at i and j . In particular every low mode does, hence $\hat{u}_i = \hat{u}_j$ exactly while $d_M(x_i, x_j) > 0$: no lower bound $A d_M$ can hold below the connectivity scale. (“Identical in every eigenvector” would be literally false—the localized twin-difference mode itself separates them—but that mode sits at the top of the spectrum, far above anything the angular map retains.) What is improvable is the *size* of θ_n : our proof takes it at the sup-norm consistency rate E_n , much larger than the bandwidth; a C^1 (gradient) consistency argument—the Lipschitz regularity theory for graph-Laplacian eigenfunctions of [24] is the natural ingredient—would control eigenvector *differences* at small separations directly and could shrink θ_n toward the connectivity scale, below which the twin obstruction is final.

Two further remarks. (1) Every constant in E_n depends on the fixed mode-count m (through $\sqrt{m}$, Φ_m , μ_1 , and the gap $\mu_{m+1} - \mu_m$); nothing here is claimed uniform in m . The uniformity in m observed empirically (§5.5) corresponds to the uniformity of (H1) [21, 22] and is exactly what

separates Theorem 8 from the full Conjecture 3; Appendix C settles its continuum side by filter (uniform for heat, scale-dependent for commute). (2) The theorem is local ($d_M \leq r_0$) because (H1) is; the global statement needs precisely the injectivity hypothesis (H3), which Corollary 9 (Step 5) carries out—and on the flat torus (H1) is itself global, so Corollary 10 (Step 6) needs no hypothesis at all.

C Uniformity in the truncation

This appendix proves Theorem 12 and Proposition 13, and records the growing-window graph-transfer conditions. Throughout, volume is normalized to 1, p_t is the heat kernel of M , and cutoffs are multiplet-closed.

Proof of Theorem 12

Let $F_t(x) = (e^{-t\mu_k/2}\varphi_k(x))_{k \geq 1} \in \ell^2$ be the *full* heat embedding with the constant mode removed; its kernel is

$$K_t(x, y) := \langle F_t(x), F_t(y) \rangle = p_t(x, y) - 1.$$

Step 1: N_t is a uniformly noncollapsing immersion for small t . For a unit tangent vector $v \in T_x M$,

$$\|dF_t(v)\|^2 = \partial_x^v \partial_y^v K_t(x, y)|_{y=x}, \quad dR_t(v) = \frac{\partial_v K_t(x, x)}{2R_t(x)}, \quad R_t^2(x) = p_t(x, x) - 1.$$

By the Minakshisundaram–Pleijel expansion (cf. [20]), $p_t(x, x) = (4\pi t)^{-d/2}(1 + t \text{scal}(x)/6 + O(t^2))$ and $\partial_x^v \partial_y^v p_t|_{y=x} = \frac{1}{2t}(4\pi t)^{-d/2}(1 + O(t))$, both uniformly on compact M . The only x -dependence of the diagonal at this order is through the curvature term, so $\partial_v p_t(x, x) = (4\pi t)^{-d/2} O(t)$, whence $dR_t(v)^2 = (4\pi t)^{-d/2} O(t^2) \cdot O(1)$, which is o of the derivative energy $\asymp t^{-1}(4\pi t)^{-d/2}$. Therefore there is $t_0(M) > 0$ such that for $t \leq t_0$ the tangential square $T_t(x, v)^2 := \|dF_t(v)\|^2 - dR_t(v)^2$ obeys

$$T_t(x, v)^2 \geq c t^{-d/2-1}, \quad R_t(x)^2 \asymp t^{-d/2}, \quad \text{hence} \quad \|dN_t(v)\|^2 = \frac{T_t(x, v)^2}{R_t(x)^2} \geq \frac{c'}{t},$$

uniformly over the unit tangent bundle (the last identity is Proposition 5).

Step 2: the truncation tail vanishes in C^2 . Eigenfunction Hölder/derivative norms grow only polynomially, $\|\varphi_k\|_{C^r} \leq C\mu_k^{(d-1)/4+r/2}$ [26], while the heat weights decay exponentially; hence $\|F_t - F_{t,m}\|_{C^2} \rightarrow 0$ as $m \rightarrow \infty$ (tail sums $\sum_{k>m} e^{-t\mu_k} \mu_k^{c(d,r)} \rightarrow 0$ by Weyl's law). Consequently $T_{t,m} \rightarrow T_t$, $R_{t,m} \rightarrow R_t$, and the C^2 -norms of $N_{t,m}$ converge, all uniformly; choose $m_0(t)$ so that for $m \geq m_0(t)$, $T_{t,m}^2 \geq \frac{1}{2}c t^{-d/2-1}$, $R_{t,m} \asymp t^{-d/4}$, and $\sup_{m \geq m_0} \|N_{t,m}\|_{C^2} =: C_2(t) < \infty$.

Step 3: from uniform immersion to uniform bi-Lipschitz. Let $a'_t := \inf_{m \geq m_0} \inf_{x, |v|=1} \|dN_{t,m}(v)\| > 0$ from Steps 1–2. For x, y with $d := d_M(x, y) \leq r_t := a'_t/(2C_2(t))$, Taylor along a minimizing geodesic γ gives

$$\|N_{t,m}(x) - N_{t,m}(y)\| \geq \|dN_{t,m}(\gamma'(0))\| d - \frac{1}{2}C_2(t) d^2 \geq \left(a'_t - \frac{1}{2}C_2(t) r_t\right) d \geq \frac{3a'_t}{4} d,$$

and the mean-value bound gives $\|N_{t,m}(x) - N_{t,m}(y)\| \leq b_t d$ with $b_t := \sup_{m \geq m_0} \text{Lip}(N_{t,m}) < \infty$. All constants depend on t and M only—not on m . $\square$

Proof of Proposition 13

Let $E_\Lambda(x, y) = \sum_{\mu_k \leq \Lambda} \varphi_k(x) \varphi_k(y)$ be the spectral function. The pointwise Weyl law with uniform remainder [26] gives $E_\Lambda(x, x) = C_d \Lambda^{d/2} + O(\Lambda^{(d-1)/2})$ with spatially constant leading coefficient, and its differentiated form (uniform C^∞ near-diagonal asymptotics of the spectral projector [27]) gives, uniformly in x and unit v ,

$$\partial_x^v \partial_y^v E_\Lambda(x, y) \big|_{y=x} = C'_d \Lambda^{d/2+1} (1 + o(1)).$$

Abel summation in the cutoff converts these into the commute-weighted sums:

$$\|dF_\Lambda(v)\|^2 = \sum_{0 < \mu_k \leq \Lambda} \mu_k^{-1} (\partial_v \varphi_k(x))^2 = c_d \Lambda^{d/2} (1 + o(1)),$$

$$G_\Lambda(x, x) := R_\Lambda^2(x) = \sum_{0 < \mu_k \leq \Lambda} \mu_k^{-1} \varphi_k(x)^2 = \begin{cases} C_d \Lambda^{d/2-1} (1 + o(1)), & d > 2, \\ C_2 \log \Lambda (1 + o(1)), & d = 2, \end{cases}$$

again uniformly in x , with spatially constant leading coefficients. Because those coefficients are constant, $\partial_v G_\Lambda(x, x)$ carries no leading term: it is $o(\Lambda^{d/2-1})$ ($d > 2$; $o(\log \Lambda)$, $d = 2$) uniformly, so the radial term $dR_\Lambda(v)^2 = (\partial_v G_\Lambda(x, x))^2 / (4G_\Lambda(x, x))$ is $o(\Lambda^{d/2})$ —lower order than the derivative energy. Hence $T_\Lambda(x, v)^2 \geq c \Lambda^{d/2}$ for $\Lambda \geq \Lambda_0$, uniformly, which is the display; dividing by R_Λ^2 gives $\|dN_\Lambda\| \gtrsim \Lambda^{1/2}$ ($d > 2$) and $\sqrt{\Lambda / \log \Lambda}$ ($d = 2$).

The S^1 computation, and the failure of fixed-scale uniformity. On S^1 ($\mu = k^2$, whole eigenspaces, weights $\mu^{-1/2} = 1/k$),

$$F_N(x) = \left(\frac{\cos kx}{k}, \frac{\sin kx}{k} \right)_{k=1}^N : \quad R_N^2 = \sum_{k \leq N} \frac{1}{k^2} \text{ (constant, bounded),} \quad \|dF_N\|^2 = \sum_{k \leq N} 1 = N,$$

so $\|dN_N\| = \|dF_N\|/R_N \asymp \sqrt{N}$ (consistent with the general formula: $\Lambda = N^2$, $d = 1$, $T_\Lambda/R \asymp \Lambda^{1/4}$). A pair at distance $\sim 1/N$ is stretched by $\sim \sqrt{N}$, so any upper Lipschitz constant valid on a *fixed* ball $\{d \leq r_0\}$ must grow like $\sqrt{m}$: no m -independent fixed-scale bi-Lipschitz theorem can hold for the commute filter, on any manifold (the same growth holds in all d). The lower bound at fixed scale, by contrast, survives on S^1 (project onto the first multiplet, whose weight is m -independent while R_N is bounded)—the failure is one-sided and lives below the wavelength $\Lambda^{-1/2}$. $\square$

Proof of Proposition 14

Plain filter. With $E_\Lambda(x, y) = \sum_{0 < \mu_k \leq \Lambda} \varphi_k(x) \varphi_k(y)$ (constant mode removed),

$$\|F_\Lambda^{(1)}(x) - F_\Lambda^{(1)}(y)\|^2 = E_\Lambda(x, x) + E_\Lambda(y, y) - 2E_\Lambda(x, y), \quad R_\Lambda^{(1)}(x)^2 = E_\Lambda(x, x).$$

The pointwise Weyl law with uniform remainder [26] gives $E_\Lambda(x, x) = C_d \Lambda^{d/2} + O(\Lambda^{(d-1)/2})$ with spatially constant leading coefficient, and for $d_M(x, y) \geq \delta$ the off-diagonal spectral function obeys $|E_\Lambda(x, y)| = O(\Lambda^{(d-1)/2})$ uniformly, oscillating in the pair separation on the wavelength scale $\Lambda^{-1/2}$ (the Bessel-type scaling limit of the spectral projector [27]). The radial difference is $R_\Lambda^{(1)}(x) - R_\Lambda^{(1)}(y) = (E_\Lambda(x, x) - E_\Lambda(y, y)) / (R_\Lambda^{(1)}(x) + R_\Lambda^{(1)}(y)) = O(\Lambda^{(d-1)/2-d/4})$, so its square is $O(\Lambda^{d/2-1})$. Feeding these into the identity of Lemma 4,

$$\|N_\Lambda^{(1)}(x) - N_\Lambda^{(1)}(y)\|^2 = \frac{2C_d \Lambda^{d/2} + O(\Lambda^{(d-1)/2})}{C_d \Lambda^{d/2} (1 + O(\Lambda^{-1/2}))} = 2 + O(\Lambda^{-1/2})$$

uniformly over separated pairs, with pair-dependence carried by $-2E_\Lambda(x, y)/E_\Lambda(x, x)$. Since the numerator kernel oscillates with sign changes on the wavelength scale, the ordering it induces has no leading-order relation to d_M : any rank statistic against geodesic distance loses its support as $\Lambda \rightarrow \infty$. This is the continuum mechanism of the plain-eigenmap collapse in the weighting ablation (§5.6).

Commute filter. The same computation with weights μ_k^{-1} replaces E_Λ by $G_\Lambda(x, y) = \sum_{0 < \mu_k \leq \Lambda} \mu_k^{-1} \varphi_k(x) \varphi_k(y)$: the diagonal is $C'_d \Lambda^{d/2-1}$ ($d > 2$; $C_2 \log \Lambda$, $d = 2$) with spatially constant leading coefficient (Step 7 of the Proposition 13 proof), the radial term is again lower order, and

$$\|N_\Lambda(x) - N_\Lambda(y)\|^2 = 2 - \frac{2G_\Lambda(x, y)}{G_\Lambda(x, x)}(1 + o(1)).$$

The correction kernel is now low-frequency dominated—its summands decay like μ^{-1} , and for $d \leq 3$ it converges off the diagonal to the Green’s function of M (for $d \geq 4$ it is the truncated Green kernel, still dominated by its low-frequency, coherent part)—a distance-structured kernel, on a homogeneous manifold a function of d_M alone. Both filters therefore concentrate at fixed scale as $\Lambda \rightarrow \infty$ (the wavelength story of Proposition 13), but the *ordering* information that rank statistics read is carried by a coherent kernel for commute and an oscillatory one for plain: the ablation’s rank separation, in the continuum. $\square$

Graph transfer at growing m , and the C^1 route

For fixed m , Appendix B is the complete transfer. For a growing spectral window $m = m(n)$, the same calibration goes through—with every constant carrying an m -subscript—whenever

$$\mathcal{E}_n(\mu_m) \gamma_m^{-1} \rightarrow 0,$$

where γ_m is the spectral gap isolating the retained block and $\mathcal{E}_n(\mu_m)$ is the graph-Laplacian consistency error on continuum eigenfunctions up to frequency μ_m ; since high-frequency eigenfunctions have polynomially large derivatives, $\mathcal{E}_n(\mu_m)$ grows polynomially in μ_m , so $m(n)$ may grow, but not too fast. The resulting cutoff is $\theta_{n,m} \gtrsim E_{n,m}/(a_m \rho_m^-)$, every factor now m -dependent. Separately, the additive pairwise error of the sup-norm argument is what forces θ_n to sit at the spectral-rate scale; a $C^{0,1}$ eigenvector theory—the Lipschitz estimates of [24] are the available ingredient—would replace it by a multiplicative derivative-level error ($\|d\hat{Z}(v)\| \geq (1 - \eta)\|dN(v)\|$ once the C^1 calibration error is below $\eta a_m \rho_m^-$), pushing the cutoff down to the connectivity scale ε_n —optimal, since below it combinatorial twins exist (Appendix B). We state these as the conditions a growing-window theorem needs, not as a theorem.

D An exploratory cross-domain probe: the moral embedding manifold

Status: exploratory. This probe sits outside the paper’s main evidence chain, for two reasons stated up front: its data path is not yet externally reproducible (the underlying embeddings are pending release with the repository, §7), and its headline number is partly a cluster-separation effect rather than local manifold structure, as decomposed below. We report it as a diagnostic-in-use, not as a supporting result for Conjecture 3.

As a probe of whether the diagnostic transfers beyond synthetic and emergent graphs, we point it at a *semantic* space: 2,400 moral/ethical text items (BGE-M3, 1024-d), stratified across six traditions, in a $k=10$ nearest-neighbour graph. The metric here is the kNN graph’s own hop geodesics,

so this measures spectral reconstructability of that graph, not external semantic validity. All-pairs angle- $\rho = 0.90$ separates it sharply from destroyed controls (per-feature shuffle 0.23, isotropic Gaussian 0.31; higher-mode $\rho = 0.01$). Two confounds keep this honest, and both are real. First, the all-pairs number is partly cluster separation—the effect row-normalization is known to enhance [14]: decomposed, *between*-tradition pairs read 0.85 but *within*-tradition pairs, the ones that speak to local manifold structure, read only 0.67 (still above the controls, well below the all-pairs figure). Second, the informative magnitude is mostly hubness: $\rho(r_i, \text{kNN in-degree}) = 0.58$, so the radius tracks node degree (Corollary 1), not a moral-specific scale. The two-factor screen of §5.3 delivers the same verdict in its own units: the moral graph sits at $\omega = +0.15$ with dimension spread 2.6—between the clean manifolds ($\omega \leq -0.50$, spread ≤ 0.51) and the destroyed substrates ($\omega \geq +0.23$, spread ≥ 0.77), so the instrument itself, not just prose, places it as an intermediate, graded case. We read the moral embedding as manifold-like *within* traditions and the cross-domain transfer as a graded diagnostic, not a headline; a generic-text control at matched n and model, k -sensitivity, and dataset documentation are deferred (§7).

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